Urban Land Subsidence Mapping using InSAR Time Series Data

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Approval

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Abstract

Urban land subsidence poses significant risks to infrastructure and development due to intensive anthropogenic activities of rapid urbanization and groundwater over-extraction, necessitating effective detection, mapping, and disaster response efforts. This thesis studies Persistent Scatterer Interferometric Synthetic Aperture Radar (PSInSAR), for urban subsidence. By using two established workflows of LiCSBAS [1] and OpenSARLab (OSL) [2], and showcasing their applications in mapping urban land subsidence, the report introduces fundamental concepts of Synthetic Aperture Radar (SAR), and PSInSAR. Presents a reproducible workflow for generating a Land Subsidence Susceptibility Model (LSSM), integrating multi-source geospatial data with time-series InSAR-derived deformation outputs from OSL MintPy's workflow. The report proposes a methodology [3], enabling robust spatial prediction and classification of subsidence-prone urban zones. Finally, the report identifies future research avenues and perspectives in the field of remote sensing based urban land subsidence prediction and monitoring.

Contents

Chapter 1

Introduction

Land subsidence (LS) is a significant and often underestimated geological hazard that is progressively causing more damage each year [4]. The rapidly growing urban areas globally are facing a significant problem of land subsidence underscoring the critical importance of detection and mapping for effective response and mitigation efforts. Ground subsidence presents major risks, particularly in densely populated regions, as it elevates risk, damages buildings and infrastructure, and leads to economic losses [5].

Land subsidence is the vertical sinking of the Earth's surface, which can happen gradually or suddenly. It can result from natural or anthropogenic activities, or a combination of both. Natural causes are typically linked to geological processes, such as tectonic activity or the compaction of sediments due to the weight of overlying materials gap [6]. The anthropogenic factors responsible for the land subsidence are caused by over-exploitation of the underground fluid such as water, petroleum, and gas [7], extraction of natural resources [8] and mineral. Case studies of urban LS in India [9], Iran [10], China [11], Mexico and Japan point to a relation between groundwater and LS. The patterns are primarily the result of groundwater extraction for domestic, industrial, and agricultural purposes, as well as sediment compaction following coastal reclamation and drainage for urban development [9].

Synthetic Aperture Radar (SAR) [12] technology has demonstrated its value in continual monitoring of urban LS, offering millimeter level precision and the ability to detect temporal deformation trends as it is crucial for early detection of subsidence, understanding the underlying mechanisms, implementing preventive measures, and forecasting future subsidence. It is important to monitor this process as it can cause severe damage in urban infrastructure such as buildings and transport network [9], and exacerbate urban pluvial floods among other disasters.

A brief survey of remote sensing technology tools for urban land subsidence indicates that PSInSAR and Small Baseline Subset (SBAS) [13] are the most widely adopted techniques, proving effective for analysing both large areas and individual buildings. Other notable algorithms like European Space Agency (ESA) extension of the interferometric capabilities of the Sentinel Application Platform (SNAP) enabling SAR interferometric time series analysis through the Stanford Method of Persistent Scatterer (StaMPS) software package [14], and Statistical-Cost, Network-Flow Algorithm for Phase Unwrapping (SNAPHU) [15], Interferometric Point Target Analysis for Deformation Mapping (IPTA) [16], and Coherent Pixels Technique Interferometry Synthetic Aperture Radar (CPT-InSAR) [17] have also been used.

Following StaMPS, several open-source software packages have emerged for InSAR processing, such as Generic Mapping Tools Synthetic Aperture Radar (GMTSAR), InSAR Scientific Computing Environment (ISCE), and others, which support interferometric mapping. Additionally, tools like Generic InSAR Analysis Toolbox (GIAnT), Miami INsar Time-series software in Python (MintPy), and Π-RATE (Poly-Interferogram Rate and Time-series Estimator) are available for displacement time series estimation. However, these tools tend to be more complex and may pose challenges for non-experts and geoscientists unfamiliar with the technical aspects.

This study applies two such platforms—OpenSARLab MintPy and LiCSBAS—to assess land subsidence across selected Indian cities. The velocity outputs generated by MintPy, both with and without atmospheric correction, are compared with those from LiCSBAS to evaluate performance and reliability. A refined MintPy velocity output is then used to generate a Land Subsidence Susceptibility Map (LSSM) [2].

By integrating SAR-derived deformation data with a spatial analytics pipeline, this research presents a robust geospatial workflow for LSSM generation. The methodology combines terrain attributes derived from Shuttle Radar Topography Mission (SRTM) data, urban Land Use Land Cover (LULC) maps from Sentinel-2 imagery, and vertical displacement from MintPy. After preprocessing and feature engineering, a classifier is trained to identify subsidence-prone zones, followed by spatial prediction and visualization.

The resulting LSSM provides actionable insights for risk mitigation and serves as a decision-support tool for urban planners, infrastructure managers, and policymakers. This integrated approach underscores the value of combining InSAR-based monitoring with scalable spatial analytics techniques, particularly in regions where traditional ground-based monitoring may be limited.

1.1 Aim and Objectives

1.1.1 Aim

Monitor urban land subsidence using InSAR techniques and develop a susceptibility model.

1.1.2 Objectives

- Map long-term vertical deformation in urban areas using Sentinel-1 SAR data.
- Compare PSInSAR (MintPy) and SBAS (LiCSBAS) velocity products for cities.
- Evaluate the effect of atmospheric corrections on velocity accuracy.
- Generate a Susceptibility Map using refined MintPy output.
- Integrate DEM, LULC, and SAR-derived deformation in a spatial analytics pipeline.
- Identify high-risk zones to inform urban planning.

1.2 Report Organization

- **Chapter 1** introduces the topic, and states the objectives and scope.
- **Chapter 2** defines key concepts, and presents a literature review.
- **Chapter 3** details the data, and study area.
- **Chapter 4** identifies the methodology and describes the workflow.
- **Chapter 5** presents the case studies, analysis, and key results.
- **Chapter 6** concludes the study and outlines policy implications and future work.

Chapter 2

Key Definitions and Literature Review

2.1 Definitions and Concepts

2.1.1 Synthetic Aperture Radar (SAR)

Synthetic Aperture Radar (SAR) is an active remote sensing system that transmits microwave signals and receives the reflected echoes to generate high-resolution imagery of the Earth's surface. Unlike optical sensors, SAR systems operate in the microwave domain, making them independent of daylight or weather conditions. The term ”synthetic” refers to the technique of simulating a large antenna aperture by moving a smaller physical antenna over time, allowing for higher resolution than would otherwise be possible from a spaceborne platform.

SAR operates on the principle of recording the time delay and phase shift of the backscattered signal. As the radar platform moves along its orbit, it emits pulses obliquely to the surface and records the reflected signals. The resolution along the direction of motion (azimuth) is improved using the Doppler effect and signal processing techniques, while the resolution in the range direction is determined by the pulse duration and bandwidth.

A key advantage of SAR is its ability to penetrate clouds and haze to acquire images in all weather conditions. The radar beam, transmitted obliquely from the satellite, captures phase and amplitude information of the reflected signal. As the platform progresses, multiple observations of the same ground target are coherently combined to form a high-resolution image.

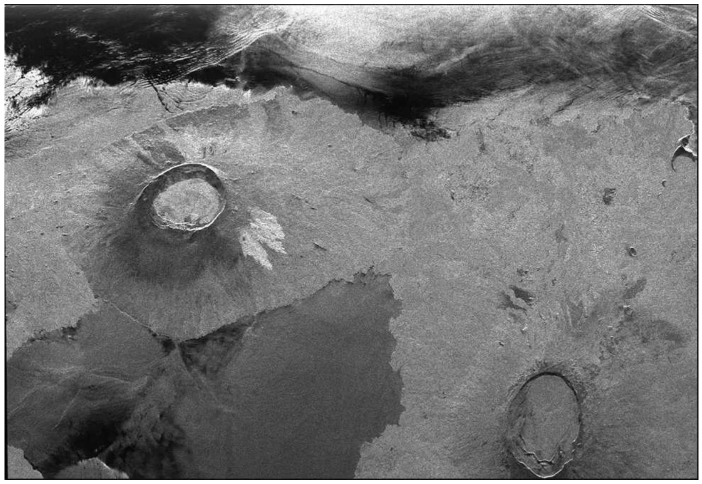


Figure 2.1: Radar image showing part of Isla Isabella in the western Galapagos Islands. This image was taken by the L-band radar in HH polarization from the Spaceborne Imaging Radar C/X-Band Synthetic Aperture Radar on the 40th orbit of the space shuttle Endeavour in 1996. Credit: NASA [18]

The radar measures the distance to each surface feature by the time delay of the returned pulse. Moreover, due to its motion, SAR exploits Doppler shifts to discriminate targets in the azimuth direction. This leads to a complex signal representing both amplitude (backscatter strength) and phase (two-way travel distance), crucial for interferometry.

2.1.2 Interferometric Synthetic Aperture Radar (InSAR)

InSAR is a remote sensing technique that uses phase differences between two or more SAR images to measure ground surface changes with centimeter to millimeter precision. The differential phase is a combination of displacement, topography, atmospheric effects, and noise:

$$\Delta\phi(\mathbf{x}) \approx \frac{4\pi}{\lambda}\,\Delta\rho(\mathbf{x}) + \Delta\phi_{\text{topo}}(\mathbf{x}) + \Delta\phi_{\text{atm}}(\mathbf{x}) + \Delta\phi_{\text{noise}}(\mathbf{x}) \tag{2.1}$$

where:

- $\mathbf{x}$ is the spatial location (pixel or coordinate) on the ground in the radar image plane.
- λ is the radar wavelength,
- $\Delta\rho$ is the line-of-sight (LOS) displacement between acquisitions,
- $\Delta\phi_{\text{topo}}$ represents topographic phase residuals due to DEM errors,
- $\Delta\phi_{\text{atm}}$ is the atmospheric phase delay difference, and

- $\Delta\phi_{\text{noise}}$ includes decorrelation and thermal noise.

Furthermore:

- $\mathbf{x}$ is a two-dimensional vector representing a position in the SAR image, either expressed in radar coordinates (range and azimuth) or in geographic coordinates (longitude and latitude).
- Each term in the interferometric phase equation is evaluated at a specific pixel $\mathbf{x}$, meaning that the total phase $\Delta\phi$ is computed on a per-pixel basis and varies spatially across the image domain.
- Therefore, $\Delta\phi(\mathbf{x})$ represents the observed phase difference at location $\mathbf{x}$ between two SAR acquisitions.

The result, called an interferogram, visualizes phase differences as fringes, each indicating a contour of equal displacement. Phase unwrapping is required to recover absolute deformation from wrapped phase measurements.

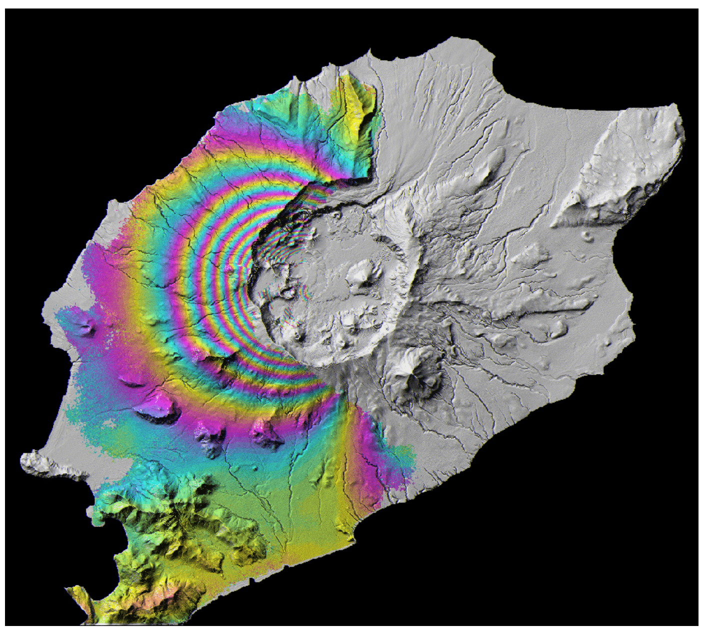


Figure 2.2: Surface deformation on Okmok, a volcano in the Aleutian Islands, is revealed in this interferometric SAR (InSAR) ERS-2 image. Each color cycle represents 2.8 cm of ground surface movement. Credit: NASA's Alaska Satellite Facility Distributed Active Archive Center (ASF DAAC) [19]

2.1.3 Introduction to Interferometric SAR (InSAR)

Interferometric Synthetic Aperture Radar (InSAR) is a powerful geodetic technique used to measure ground surface displacement by comparing radar images of the same area acquired at different times. These comparisons produce **interferograms**, which reveal subtle changes in Earth's surface caused by phenomena such as earthquakes, volcanic activity, and land subsidence.

InSAR can be categorized into two primary modes:

- **Single-Pass InSAR:** Utilizes two antennas mounted on the same platform (e.g., airborne or shuttle-based sensors) to acquire simultaneous radar images. This configuration eliminates temporal decorrelation.
- **Repeat-Pass InSAR:** Uses radar images acquired from the same satellite position at different times. This mode is common in satellite-based SAR missions and is subject to temporal and geometric decorrelation.

The phase of a radar signal encodes the two-way distance between the satellite and the ground. By comparing the phase values between two SAR acquisitions, small changes in surface elevation or deformation can be detected. The **baseline**, or spatial separation between acquisition positions, plays a crucial role:

- *Short baselines* are preferable for deformation analysis.
- *Long baselines* enhance sensitivity to topography and are typically used for DEM generation.

An **interferogram** is a phase image that visualizes the phase difference between two co-registered SAR acquisitions. These phase differences appear as fringes—concentric bands representing areas of constant phase shift. Each fringe corresponds to a 2π phase cycle, which, depending on radar wavelength, may indicate a certain amount of vertical or horizontal displacement.

Key interpretations from fringe patterns:

- *Tightly spaced fringes* imply rapid or large ground displacement.
- *Widely spaced fringes* suggest gradual or small changes.
- *Colour or greyscale variation* denotes different levels of surface motion or elevation.

Note that interferograms must be interpreted carefully, as the phase signal is also affected by atmospheric effects (e.g., water vapour variability) and residual topographic errors. Proper correction techniques such as atmospheric filtering and DEM refinement are essential to isolate true deformation signals.

2.1.4 Limitations of Traditional InSAR

Despite its effectiveness, traditional Interferometric Synthetic Aperture Radar (InSAR) techniques face several limitations that have motivated the development of more sophisticated approaches, such as time-series InSAR analysis. This report aims to highlight and demonstrate the advantages of such time-series methods.

Traditional InSAR typically relies on a limited number of interferograms—often derived from only two SAR acquisitions—which restricts the depth and reliability of deformation analysis. This is especially suboptimal when high-frequency datasets, such as those from the Sentinel-1 mission, offer repeated coverage of the same area at regular intervals. Failing to leverage the full archive of available images results in reduced temporal resolution and accuracy. Time-series InSAR analysis addresses this shortcoming by using the complete stack of SAR acquisitions, thereby enhancing the precision of displacement measurements.

Specific limitations of traditional InSAR include:

1. *Line-of-Sight (LOS) Sensitivity:* InSAR measures displacement only along the radar's line of sight. While it is sensitive to vertical and some components of horizontal motion, it cannot detect movement that is perpendicular to the radar beam. For example, in the case of strike-slip faults like the San Andreas Fault, motion that occurs parallel to the satellite's path may remain undetected depending on the viewing geometry.

2. *Temporal Coherence Loss:* The accuracy of InSAR measurements relies on consistent surface scattering properties over time. Natural processes such as vegetation growth, erosion, and human activity can change the surface characteristics, leading to temporal decorrelation. This limits the maximum temporal baseline and reduces the reliability of measurements over longer intervals.

3. *Atmospheric Delays:* Atmospheric disturbances, particularly variations in water vapor, can introduce phase delays that are unrelated to ground deformation. These delays affect the radar signal's travel time, introducing noise into the interferogram. Distinguishing atmospheric phase components from actual ground displacement remains a significant challenge. In some documented cases, features initially interpreted as deformation due to volcanic activity were later identified as atmospheric artifacts.

4. *Residual Topographic Errors:* Even after applying a Digital Elevation Model (DEM) to remove the topographic phase component, inaccuracies in the DEM can introduce residual errors. While these errors are generally small, they

can significantly affect displacement estimates when measurements aim for millimeter-scale precision over long periods.

5. *Orbital Errors:* Although modern satellites maintain highly precise orbital trajectories, small inaccuracies in the satellite's position can introduce phase ramps across the interferogram. These artifacts, if uncorrected, can bias the interpretation of displacement patterns, particularly in large-area studies.

2.1.5 Differential Interferometric Synthetic Aperture Radar (DInSAR)

Differential Interferometric Synthetic Aperture Radar (DInSAR) is a technique that isolates surface deformation signals by removing the topographic phase component from standard interferometric SAR (InSAR) measurements using a reference Digital Elevation Model (DEM).

Consider a single ground target point P. The satellite acquires a first SAR image from position M, measuring a phase:

$$\phi_M = \frac{4\pi}{\lambda} \cdot MP + \phi_{\text{scatt}} \tag{2.2}$$

where MP is the sensor-to-target distance, λ is the radar wavelength, and ϕ_{scatt} represents the phase shift generated by interaction with the target.

A second image is acquired from position S, yielding:

$$\phi_S = \frac{4\pi}{\lambda} \cdot SP + \phi'_{\text{scatt}} \tag{2.3}$$

The interferometric phase is then:

$$\Delta\phi = \phi_S - \phi_M = \frac{4\pi}{\lambda}(SP - MP) \tag{2.4}$$

This phase difference is related to the distance difference $SP - MP$ and contains contributions from surface deformation, topography, atmosphere, orbit errors, and noise.

To isolate the deformation signal, DInSAR subtracts the topographic contribution using an external DEM, yielding the simplified observation model:

$$\Delta\phi = \phi_{\text{disp}} + \phi_{\text{res}} + \phi_{\text{atm}} + \phi_{\text{orb}} + \phi_{\text{noise}} + 2\pi k \tag{2.5}$$

where:

- ϕ_{disp} — phase due to surface displacement,
- ϕ_{res} — residual topographic error (RTE),
- ϕ_{atm} — atmospheric phase delay,
- ϕ_{orb} — phase due to orbital inaccuracies,
- ϕ_{noise} — decorrelation and thermal noise,
- $2\pi k$ — integer phase ambiguity due to phase wrapping.

The objective of DInSAR is to retrieve ϕ_{disp} by minimizing or correcting for the other components. This is achievable in areas with strong coherence, typically observed in:

- **Permanent Scatterers (PS):** Targets (e.g., buildings, rocks) that consistently reflect radar signals with high phase stability.
- **Distributed Scatterers (DS):** Areas with homogeneous scattering (e.g., bare soil or dry grasslands) that maintain relative phase consistency over time.

Limitations of DInSAR:

1. *Temporal and Geometric Decorrelation:* Reduces coherence between acquisitions, especially over vegetated or water-covered areas.
2. *Phase Unwrapping Errors:* Estimating the correct integer k is difficult in low-coherence or noisy regions.
3. *Atmospheric Effects:* Spatially correlated atmospheric delays can mask or distort deformation signals if not properly corrected.

Despite these challenges, DInSAR remains a foundational technique for identifying surface displacement, especially in relatively stable environments or when used as a precursor to more advanced time-series methods like PSInSAR or SBAS.

2.1.6 Persistent Scatterer InSAR (PSInSAR)

PSInSAR is a multi-temporal InSAR technique that identifies stable reflectors (Persistent Scatterers, PS) and tracks their phase evolution over time. Traditional Differential InSAR (DInSAR) methods often fail in vegetated or rapidly changing surfaces due to temporal and geometric decorrelation. PSInSAR, introduced by Alessandro Ferretti in 2001 [20], overcomes these limitations by focusing on coherent, stable reflectors known as Persistent Scatterers (PS). These are typically man-made structures or rocks that retain phase stability over long time spans.

The modeled phase at a PS target i is:

$$\phi_i(t) = \frac{4\pi}{\lambda} d_i(t) + \frac{4\pi}{\lambda} \frac{B_\perp(t)}{r \sin\theta} \Delta h_i + \phi_i^{APS}(t) + \phi_i^{\text{noise}}(t) \tag{2.6}$$

where:

- $d_i(t)$ is the line-of-sight (LOS) deformation at time t,
- Δh_i is the residual DEM error for the persistent scatterer i,
- $B_\perp(t)$ is the perpendicular baseline at time t, and
- $\phi_i^{APS}(t)$ is the atmospheric phase screen component.

By analysing a large stack of interferograms, PSInSAR separates deformation signals from atmospheric noise and DEM errors, enabling estimation of linear or non-linear deformation time series at millimetre accuracy.

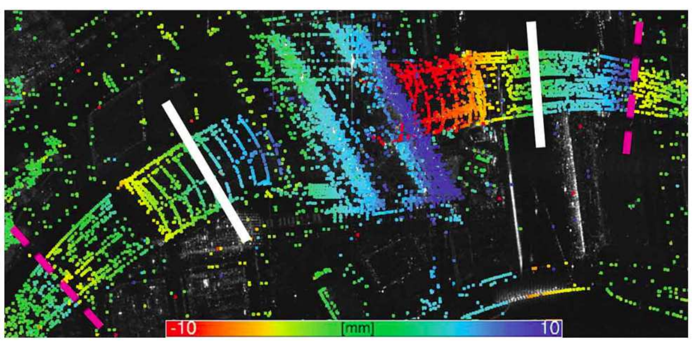


Figure 2.3: Seasonal deformation at Berlin Central Station detected using Persistent Scatterer InSAR (PSInSAR) [21]

Key Advantages of PSInSAR

1. *High Precision:* PSInSAR achieves millimetre level accuracy in deformation monitoring, particularly effective in urban environments.

2. *Atmospheric Correction:* Temporal averaging allows separation of atmospheric noise from deformation signals.

3. *DEM Refinement:* DEM errors are corrected through baseline and phase modeling.
4. *Long-Term Monitoring:* Suitable for multi-year analysis of slow processes like subsidence.

Persistent Scatterer Interferometry (PSI) represents a specific class of Differential InSAR (DInSAR) techniques that exploits multiple SAR images acquired over the same area, along with appropriate data processing and analysis procedures, to isolate the deformation phase component from other contributors.

The main outcomes of a PSI analysis include:

- the **deformation time series**, and
- the **deformation velocity** estimated over the analysed Permanent Scatterers (PSs) or Distributed Scatterers (DSs).

In the remainder of this document, the term *PSs* will be used to refer to both Permanent and Distributed Scatterers, unless otherwise specified.

Another important outcome of PSI is the so-called **residual topographic error (RTE)**, which represents the difference between the actual height of the scattering phase centre of a given PS and the height derived from the Digital Elevation Model (DEM) at the same location. The RTE is a critical parameter for achieving accurate geocoding of PS targets.

2.1.7 Overview of Persistent Scatterer InSAR (PSInSAR)

Persistent Scatterer InSAR (PSInSAR) is an advanced radar interferometry technique designed to enhance the accuracy and reliability of ground deformation measurements, especially in urban or densely built environments. It identifies and monitors **Persistent Scatterers (PS)**, which are stable, highly reflective targets, such as buildings, bridges, rock outcrops, and critical infrastructure that exhibit consistent radar backscatter over long time periods in a SAR image series.

By analysing these PS points across multiple SAR acquisitions, PSInSAR enables high-precision measurements of surface displacement, often achieving millimetre level accuracy. This is accomplished by minimizing phase decorrelation effects caused by environmental factors such as vegetation growth, soil moisture variation, or seasonal changes.

PSInSAR relies on a dense time series of SAR observations, often spanning several years. The technique is particularly effective for detecting *slow-moving* ground deformation phenomena such as subsidence, gradual slope movement, and tectonic strain accumulation. The extended temporal coverage also facilitates the separation and filtering of atmospheric delays, leading to improved phase accuracy and cleaner deformation signals.

A key strength of PSInSAR lies in its ability to generate a **displacement time series** for each detected PS target. These time series provide valuable insights into the temporal dynamics of deformation—enabling detection of continuous, episodic, or accelerating motion trends. For instance, PSInSAR is commonly used to monitor long-term land subsidence caused by groundwater extraction or to assess infrastructure vulnerability over time.

PSInSAR has a broad spectrum of applications:

1. *Urban Monitoring:* Detection of structural deformation, building settlement, or subsidence in densely populated areas.
2. *Geohazard Assessment:* Monitoring of landslides, earthquakes, volcanic activity, and mining-induced ground motion.
3. *Infrastructure Stability:* Surveillance of critical assets such as dams, bridges, highways, tunnels, and pipelines.
4. *Environmental Change Detection:* Analysis of long-term processes such as aquifer depletion, tectonic uplift, or compaction of sedimentary basins.

The ability of PSInSAR to combine high spatial resolution with long-term temporal continuity makes it a vital tool for geospatial monitoring, infrastructure management, and early warning systems in hazard-prone regions.

2.2 Literature Review

2.2.1 Early Development of InSAR and PSInSAR

The inception of InSAR for geophysical applications began in the 1990s with the advent of satellite-borne SAR missions like ERS-1/2. Early work [22] demonstrated surface deformation detection using repeat-pass SAR data. However, issues like temporal decorrelation and atmospheric delays limited its applicability.

Ferretti [20] revolutionized the field by introducing the PSInSAR method. This allowed the use of long temporal stacks of SAR data to extract stable reflectors and estimate deformation rates with unprecedented accuracy. Their method laid the foundation for commercial PSInSAR services and operational deformation monitoring.

2.2.2 Advancements in Multi-Temporal InSAR

Subsequent studies extended PSInSAR by incorporating spatial coherence networks (e.g., StaMPS [23]), enhancing robustness in non-urban and semi-vegetated areas. SBAS (Small Baseline Subset) [24] was developed to reduce spatial decorrelation by limiting interferogram baseline lengths.

Recent methods integrate time series inversion techniques, atmospheric filtering, and non-linear model fitting. For instance volcano deformation [25] and infrastructure monitoring [26].

2.2.3 InSAR for Land Subsidence in Urban India

Studies have increasingly applied InSAR to map urban land subsidence in Indian cities. Sentinel-1 data between 2014 and 2020 was used to monitor deformation in Delhi and found clear links to groundwater over-extraction [27]. Similarly, a combined PSInSAR and hydrogeological datasets mapped sinking hotspots in Lucknow [28].

InSAR with GPS frameworks analysed deformation trends in Kolkata, confirming correlations with aquifer depletion zones [29]. These studies highlight the growing relevance of open-source InSAR platforms for large-scale urban monitoring.

2.2.4 Limitations and Research Gaps

Despite its promise, PSInSAR is limited by:

- **Sparse PS Density:** In rural or vegetated areas, PS density is low, reducing spatial resolution.
- **Reference Point Bias:** All deformation is relative to a selected reference, which may influence interpretation.
- **Atmospheric Residuals:** Although atmospheric phase (APS) is estimated, some spatially correlated noise remains.

Recent work focuses on fusing InSAR with GNSS, hydrological, and machine learning models to overcome these limitations and enhance interpretability.

Chapter 3

Dataset and Study Area

3.1 Datasets

The ability to monitor surface deformation over time using Persistent Scatterer Interferometry (PSI) depends critically on the availability and density of Synthetic Aperture Radar (SAR) acquisitions. This availability is governed by several factors, including:

- the operational status of active SAR sensors,
- satellite revisit intervals,
- data acquisition policies (e.g., routine versus on-demand modes), and
- orbital parameters such as permissible perpendicular baselines.

Currently, several SAR missions such as *COSMO-SkyMed*, *TerraSAR-X*, *Sentinel-1*, and *ALOS-2* are actively acquiring data. However, their acquisition strategies differ across regions and applications, affecting global PSI applicability.

PSI typically requires a minimum of 15–20 SAR images for reliable deformation analysis in the C-band. In contrast, X-band datasets, owing to their higher spatial resolution and shorter wavelength, can sometimes produce meaningful results with fewer images. Nonetheless, an increased number of SAR acquisitions generally improves the accuracy of both deformation velocity maps and time series, especially when applying advanced PSI modeling techniques.

This project integrates a multi-source geospatial dataset stack for mapping and analyzing urban land subsidence and generating a Land Subsidence Susceptibility Map (LSSM). The datasets span satellite-derived SAR and optical imagery, digital elevation models, hydrological baselayers, and vector geospatial boundaries.

3.1.1 Sentinel-1 SAR Data :

Source: Copernicus Open Access Hub / Alaska Satellite Facility (ASF)

Type: Sentinel-1 Level-1 SLC, IW mode, VV and VV+VH polarization

Purpose: Input for InSAR time-series analysis using MintPy to detect millimeter-scale vertical land motion, with ERA5-based atmospheric correction in MintPy.

The Sentinel-1 Synthetic Aperture Radar (SAR) data used in this study were sourced from the Copernicus Open Access Hub and the Alaska Satellite Facility (ASF). The acquisitions were collected in Single Look Complex (SLC) Level-1 format, Interferometric Wide (IW) swath mode, which is optimized for land-based interferometry applications. Each SLC scene contains dual polarization data (VV and VH). For the purposes of InSAR time-series analysis, the VV polarization was primarily used due to its higher phase coherence and signal stability in urban environments, while VV+VH polarization was retained in the dataset for potential multi-channel analysis or classification tasks. The Sentinel-1 SLC data serves as the primary input for interferometric time-series analysis conducted using OpenSARLab MintPy and LiCSBAS. The MintPy pipeline employs the smallbaselineApp.py workflow for generating Small Baseline Subset (SBAS) interferograms and displacement time series.
To mitigate atmospheric phase delays, the processing includes tropospheric correction using ERA5 weather reanalysis data retrieved via the Copernicus Climate Data Store API. Leveraging the high temporal revisit frequency (6–12 days) and spatial resolution of Sentinel-1, the workflow enables millimeter-level precision in monitoring subtle vertical land movements, making it highly effective for tracking urban land subsidence dynamics.

3.1.2 Sentinel-2 Optical Imagery :

The Sentinel-2 optical imagery used in this study was obtained from the ESA Copernicus program, specifically the S2 MSIL2A product, which provides Level-2A atmospherically corrected surface reflectance data from the Multispectral Instrument (MSI). This data was used to generate Land Use Land Cover (LULC) maps through supervised classification using a Random Forest algorithm in R, with training samples prepared using the Semi-Automatic Classification Plugin (SCP) in QGIS. The resulting LULC layers were co-registered with MintPy derived vertical velocity rasters to extract urban land characteristics such as built-up, vegetation, and water bodies that influence subsidence susceptibility across the study area.

3.1.3 SRTM Digital Elevation Model (DEM) :

The Digital Elevation Model (DEM) used in this study is derived from NASA's Shuttle Radar Topography Mission (SRTM) at a 1 arc second (approximately 30 m) resolution. Terrain attributes were computed from the DEM, including slope, aspect, curvature, Topographic Wetness Index (TWI), and flow accumulation. These surface features provide critical insight into hydrological behaviour, surface morphology, and geomorphic stress; important factors in assessing land subsidence susceptibility, particularly in areas affected by groundwater movement and structural instability.

3.1.4 MintPy Velocity Outputs :

The vertical deformation data used in this study were generated using OpenSARLab, a cloud-based InSAR processing environment hosted on NASA JupyterHub operated by the Alaska Satellite Facility. The primary output, stored in the velocity.h5 time-series file produced by MintPy, contains vertical velocity maps derived from Sentinel-1 SAR data using a Small Baseline Subset (SBAS) approach. These velocity maps processed both with and without atmospheric correction using ERA5 reanalysis data serve as an inputs to the Land Subsidence Susceptibility Model. Specifically, they are used both as reference indicators of observed deformation and as target labels for training the classification model.

3.1.5 LiCSBAS Displacement Products :

The LiCSBAS-based deformation products used in this study were generated using the LiCSAR and LiCSBAS processing framework, which enables automated InSAR time-series analysis from Sentinel-1 data. Outputs were produced using both the default pipeline and an atmospheric correction-enhanced version incorporating Generic Atmospheric Correction Online Service (GACOS) data. These results were used to validate and cross-compare with MintPy-derived velocity maps, helping to assess the consistency and reliability of different InSAR processing approaches. Additionally, the LiCSBAS outputs informed the selection of reference velocity thresholds for subsidence classification.

3.1.6 Vector and Ancillary Data :

A variety of vector and ancillary datasets were integrated into the analysis to support spatial preprocessing. Administrative boundary shapefiles delineating urban extents were used to clip and constrain raster datasets to the study area. Infrastructure layers, such as road networks and built-up zones, were extracted from OpenStreetMap (OSM) to assist in identifying dense urban features relevant to subsidence patterns. Hydrological context was provided by global groundwater and surface water datasets, including those from UN IGRAC and the JRC Global Surface Water repository, enabling assessment of water-related drivers of deformation. For model training and validation, stratified random sampling points were generated and labeled using the Semi-Automatic Classification Plugin (SCP) in QGIS, ensuring spatially representative input for supervised classification and accuracy evaluation.

3.2 Study Area

The study focuses on rapidly urbanizing cities in India; **Lucknow, Kolkata, Delhi, Chandigarh, and Mumbai**; which are known to be land subsidence prone due to a combination of urban expansion, groundwater over extraction, and alluvial geomorphology.

3.2.1 Geographical Characteristics

- **Lucknow:** A growing Gangetic plains city with extensive aquifer depletion.
- **Kolkata:** A coastal megacity affected by sediment compaction, and pluvial flood.
- **Delhi:** India's capital, with dense infrastructure and high groundwater extraction.
- **Chandigarh:** A planned city with increasing localized aquifer stress.
- **Mumbai:** A reclaimed coastal metropolis with subsurface instability and tidal dynamics.

3.2.2 Data Spatial Coverage

All datasets used in this study are spatially aligned and clipped to the administrative boundaries of the cities under investigation. The temporal coverage of the Sentinel-1 SAR data ranges from early 2019 to mid-2025, capturing seasonal and anthropogenic variations in surface displacement.

This spatial diversity enables cross-comparison across different hydrogeological and urban typologies, strengthening the generalization of the LSSM modeling approach.

Chapter 4

Methodology

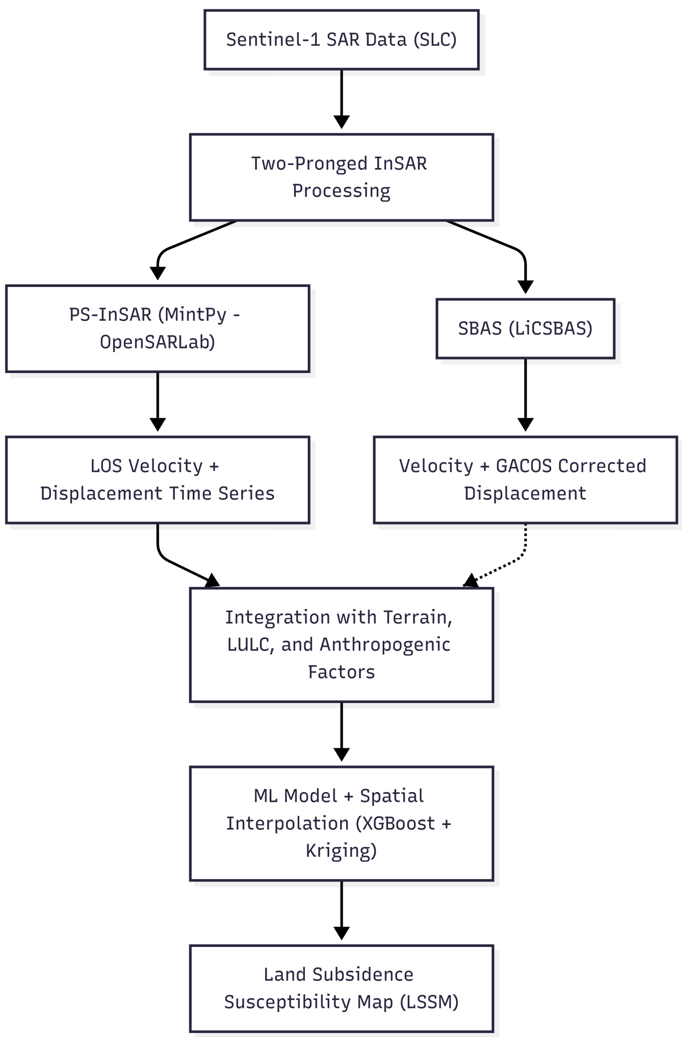


Figure 4.1: Integrated workflow for land subsidence susceptibility modeling using InSAR and ML

This study adopts a two-pronged methodological approach for land subsidence mapping over am urban area using Sentinel-1 SAR data. The first step is to obtain ground deformation through multi-temporal InSAR analysis employing two methods:

1. *Persistent Scatterer InSAR (PS-InSAR):* through the MintPy toolbox on OpenSARLab and

2. *Small Baseline Subset InSAR (SBAS):* employing LiCSBAS.

These procedures provide spatially continuous measurements of land subsidence, offering valuable insight into deformation patterns over time. Subsequently, the Persistent Scatterer InSAR (PS-InSAR) product is integrated with geo-environmental variables using a machine learning pipeline implemented in R. This pipeline comprises a modular series of R scripts designed to handle data integration, feature selection, model training, validation, and the generation of a Land Subsidence Susceptibility Map (LSSM). A precursor tp the analysis, a Differential InSAR (DInSAR) displacement map is generated to illustrate the advantages of time series based mapping.

4.1 DInSAR Workflow in SNAP

The Differential InSAR (DInSAR) processing workflow implemented in ESA's SNAP toolbox is illustrated in Figure 4.2. This workflow allows the extraction of surface displacement information from a pair of Sentinel-1 SLC images acquired at different times. The key steps include preprocessing, interferogram generation, topographic phase removal, filtering, phase unwrapping, and conversion to displacement.
The process begins with the ingestion of Sentinel-1 SLC products (master and slave), followed by orbit file application and TOPSAR splitting to isolate the relevant sub-swath and burst. The images are then co-registered using back-geocoding and enhanced spectral diversity techniques.
An interferogram is generated and deburst to create a seamless phase product, which is subsequently topographically corrected using an external DEM. Multi-looking (8×2) is applied to reduce speckle and improve coherence. Goldstein phase filtering enhances phase quality for unwrapping.
The filtered interferogram is exported to SNAPHU for phase unwrapping, and the unwrapped phase is re-imported into SNAP. Finally, the unwrapped phase is converted into displacement using the radar wavelength and look angle, resulting in a geocoded displacement map. This map serves as a preliminary deformation layer to identify subsidence zones for further time-series analysis. The DInSAR workflow is described in Figure 4.2.

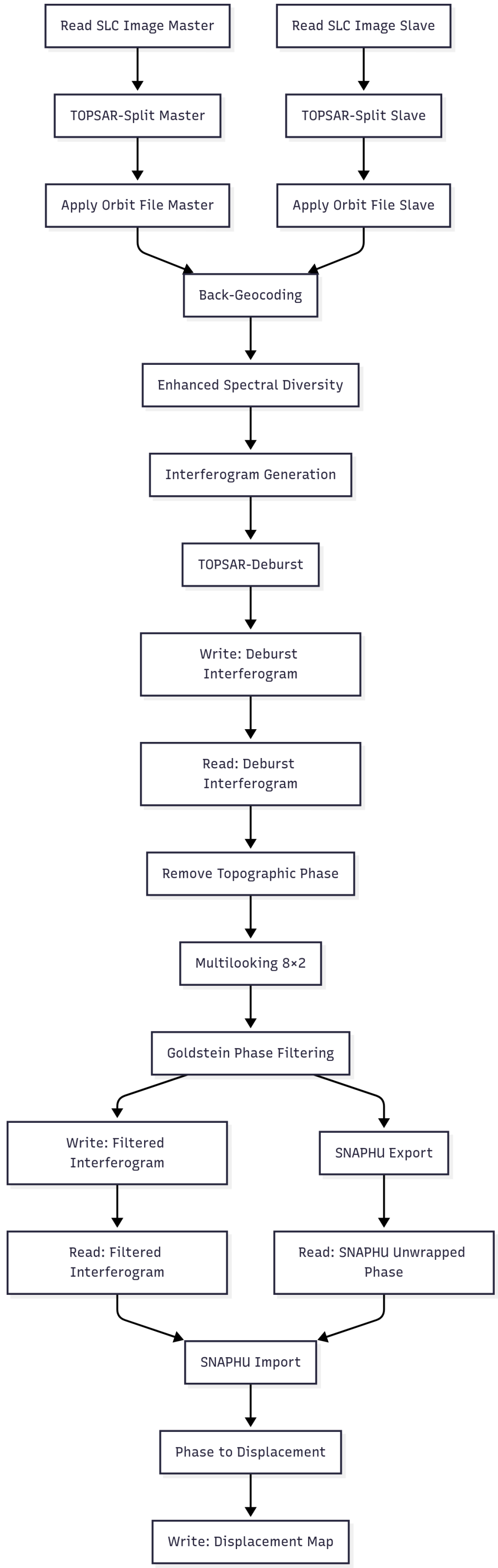


Figure 4.2: DInSAR workflow in SNAP

4.2 InSAR Time Series with MintPy (OpenSARLab)

Sentinel-1 Level-1 SLC (IW mode, VV or VV+VH polarization) are processed using the MintPy SBAS workflow in OpenSARLab. After interferogram generation via ASF HyP3, MintPy takes the unwrapped stack to estimate displacement time series and mean LOS velocity. The process involves interferogram network optimization, atmospheric filtering (via spatial-temporal filters), ramp correction, and velocity derivation. A reference pixel is selected in a stable area (example Delhi Ridge). The result is a GeoTIFF raster of LOS velocity representing average vertical displacement, which is used for training the susceptibility model. The workflow is described in Figure 4.3. While the PSInSAR workflow is described in Figure 4.4.

4.3 SBAS Time Series with LiCSBAS

LiCSBAS is used on pre-processed LiCSAR interferograms for the same frame to complement and validate MintPy outputs. GACOS atmospheric corrections are applied. The SBAS inversion, based on loop-closed interferogram networks, produced velocity and displacement time series on a 90 m grid. The LiCSBAS outputs showed spatial agreement with MintPy-derived hotspots. Though not used directly in modeling, this cross-validation adds robustness to the detection of subsidence patterns. The LiCSBAS workflow is described in Figure 4.5.

4.4 Susceptibility Modeling

Subsidence data derived from MintPy was combined with conditioning factors to simulate susceptibility with seven successive scripts organized in R for the procedure.The workflow integrates PSInSAR deformation data with terrain, land cover, and anthropogenic variables to model land subsidence susceptibility. Using features like slope, TWI, LULC, and construction density, a Machine Learning model is trained and evaluated. The resulting susceptibility map, is classified into five risk zones, and validated using PS-InSAR products. This workflow provides a scalable method for subsidence risk assessment using remote sensing and machine learning.

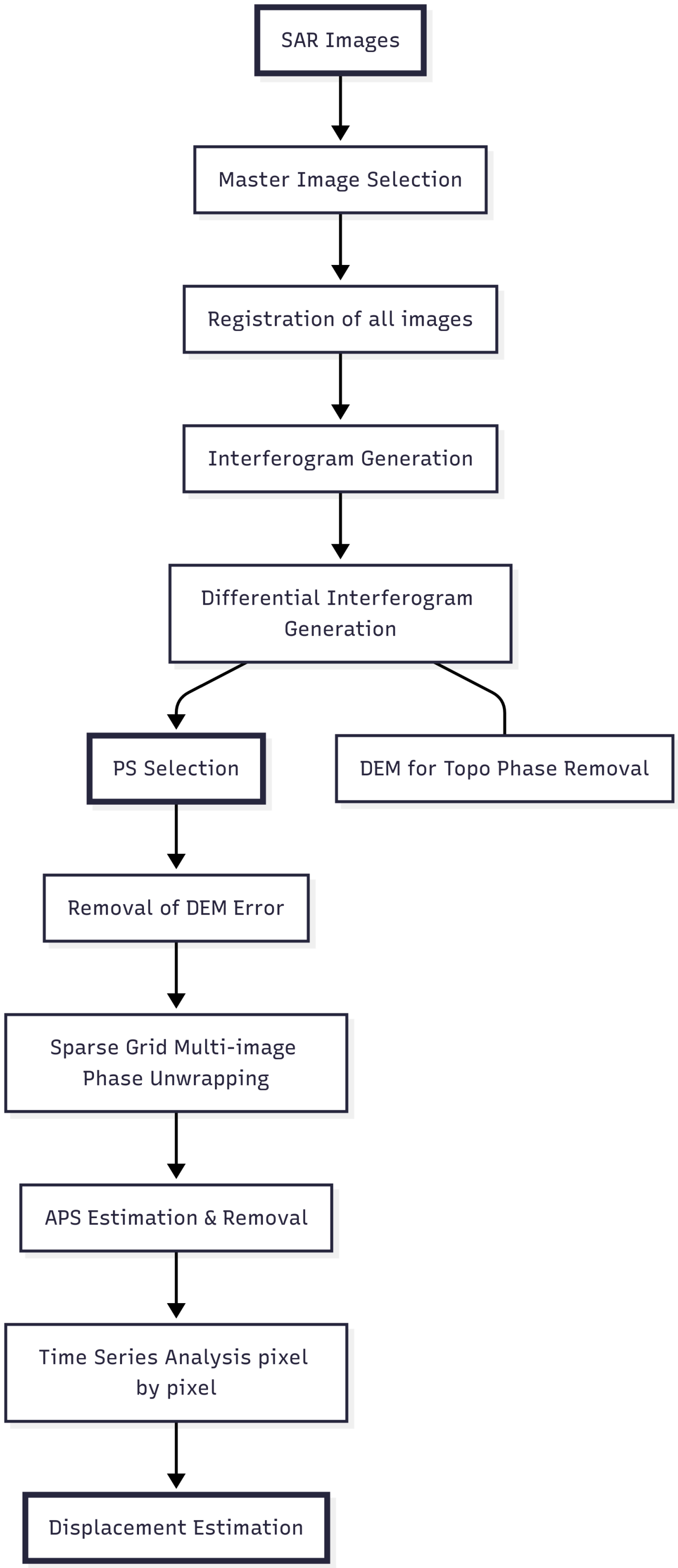


Figure 4.3: PSInSAR Standard Workflow

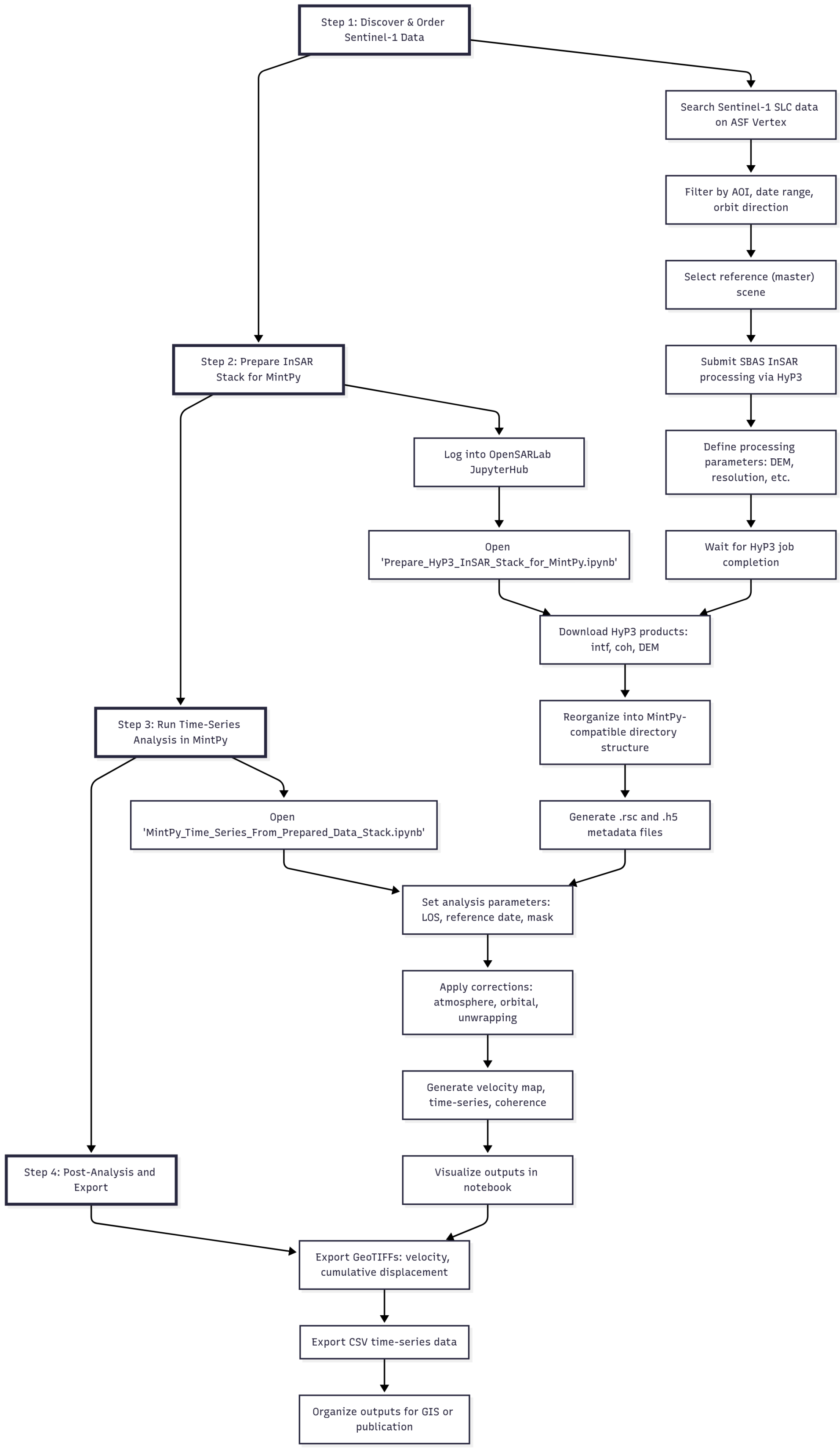


Figure 4.4: InSAR Standard Workflow in MintPy OpenSARLab

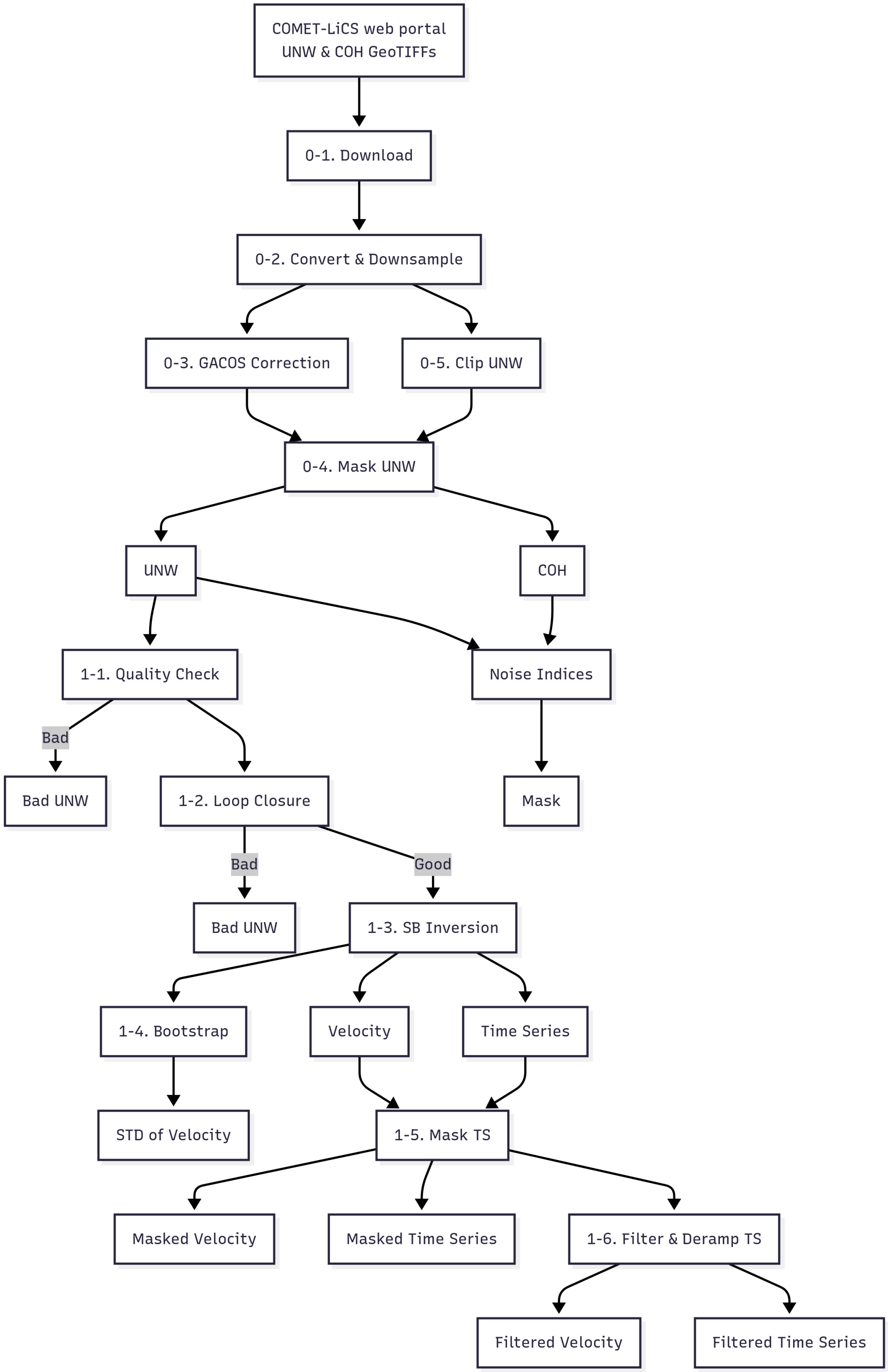


Figure 4.5: LiCSBAS Workflow

4.5 R Scripts Used for LSSM :

1. **Script 1:** Extract terrain attributes like elevation, slope, Stream Power Index (SPI), and Topographic Wetness Index (TWI), etc. from the SRTM Digital Elevation Model (DEM). These variables are assessed for multicollinearity and subsequently converted from continuous to categorical values for use in susceptibility modeling.

2. **Script 2:** Classified and validated Land Use Land Cover (LULC) categories using a Random Forest model with spatially stratified training and testing datasets. Key anthropogenic variables such as road density and construction areas are incorporated due to their known influence on land subsidence.

3. **Script 3:** Integrated the classified LULC raster with PS-InSAR-derived deformation data to generate labeled training samples. A velocity threshold of -10 mm/year is applied to Sentinel-1-derived `velocity.h5` data, which is then merged with terrain attributes to build a composite feature set.

4. **Script 4:** Conduct data preprocessing including synthetic sample generation, normalization, and filtering. Redundant features are removed using Pearson Correlation Coefficients (PCC) and Variance Inflation Factor (VIF) analysis to mitigate multicollinearity.

5. **Script 5:** Develop and evaluate machine learning models for subsidence susceptibility classification. Model performance are assessed using AUC, F1-score, recall, and precision metrics. Quantify feature importance, highlighting key terrain and anthropogenic drivers.

6. **Script 6:** Apply the trained model to the entire study area to produce a continuous Land Subsidence Susceptibility Map (LSSM) via Ordinary Kriging after selecting an appropriate variogram model . The map is categorized into five risk zones (Very Low to Very High) using quantile breaks. Outputs were scaled and exported as GeoTIFF rasters.

7. **Script 7:** Visualize PS-InSAR outputs from MintPy, including cumulative displacement, line-of-sight (LOS) velocity, and coherence rasters. Validated PS points were used directly, without interpolation to confirm spatial integrity prior to model training.

Chapter 5

Results

This chapter presents the results obtained from the two-pronged InSAR processing and the machine learning-based subsidence susceptibility modeling workflow. The outputs are discussed in three parts:

1. DInSAR Derived displacement patterns
2. Results from OpenSARlab MintPy
3. SBAS based validation from LiCSBAS
4. Land Subsidence Susceptibility Map (LSSM)

Visualizations and quantitative analyses are included to interpret spatial trends and model performance.

5.1 DInSAR Derived displacement patterns

The initial DInSAR based assessment using SNAP served as a baseline deformation indicator. The generated interferograms revealed localized vertical displacement across the Lucknow region. The full swath analysis (Figure 5.1) shows deformation signals potentially related to atmospheric noise and topographic phase artefacts, common in single-pair interferometry.

When clipped to a 30 × 30 km urban extent (Figure 5.2), more coherent displacement fringes emerged, with clearer differentiation of deformation zones. However, due to limitations such as temporal decorrelation and uncorrected atmospheric delays, DInSAR results were treated as qualitative indicators rather than inputs for quantitative modeling.

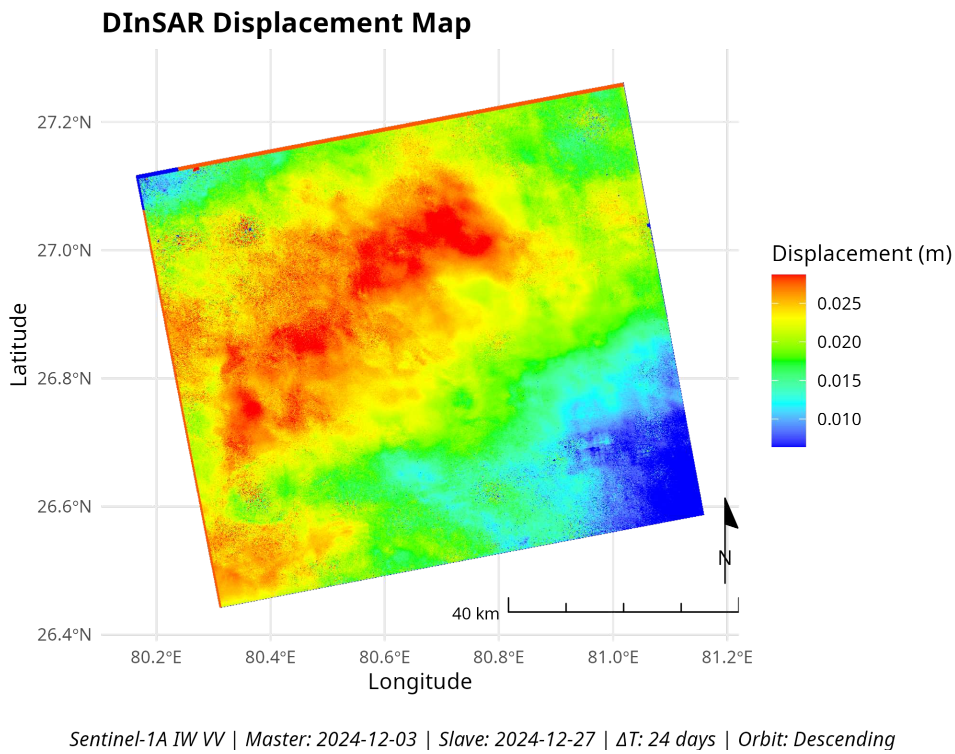


Figure 5.1: Displacement over Lucknow derived from DInSAR (SNAP) – entire swath

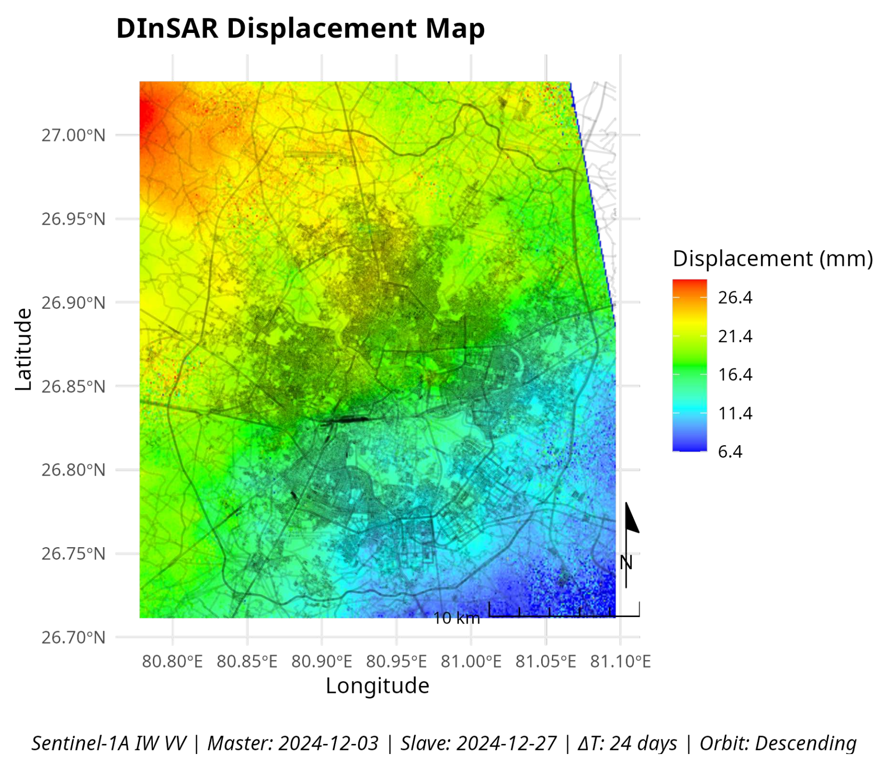


Figure 5.2: Displacement over Lucknow derived clipped to urban core (30 × 30 km) derived from DInSAR (SNAP)

5.2 OpenSARlab MintPy Derived Subsidence Patterns

The PSInSAR analysis on Sentinel-1 SLC data over the Lucknow region provided millimetre scale deformation patterns (Figure 5.3 and Figure 5.4) with deformation rates exceeding -25 mm/year reaching -63.9 mm/year in the peri-urban areas.

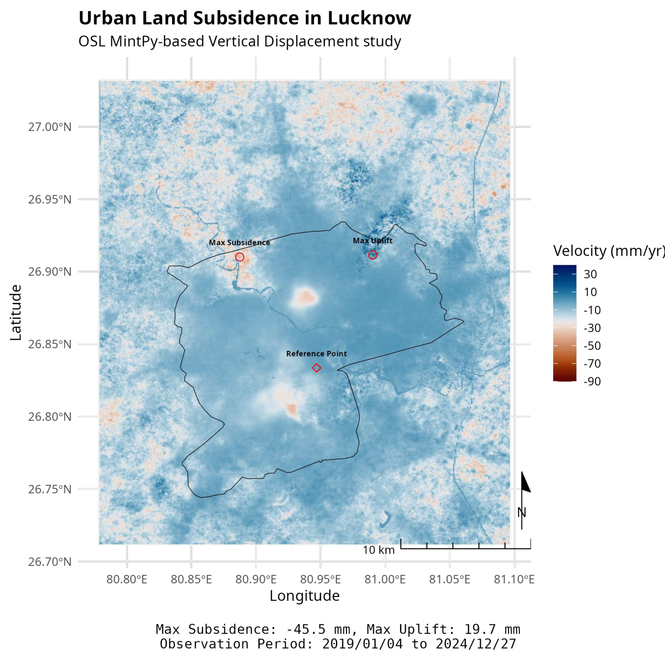


Figure 5.3: OSL MintPy derived mean LOS Velocity over Lucknow city (ΔT: 5 years)

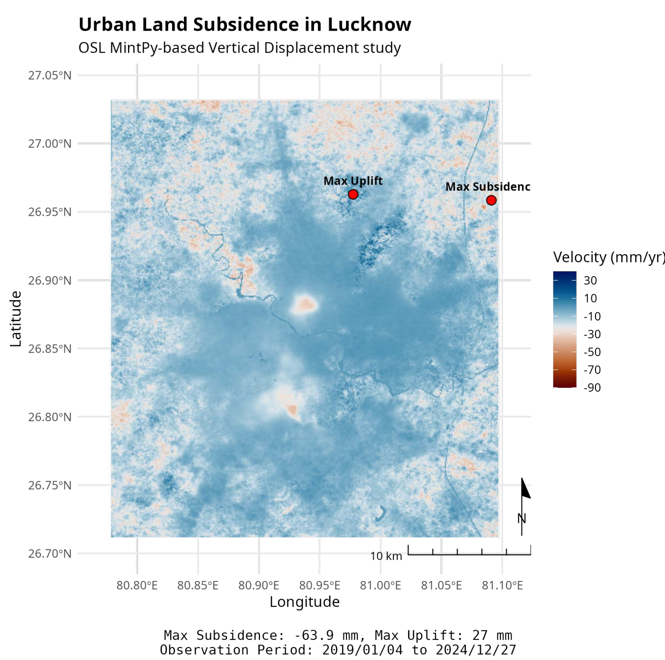


Figure 5.4: OSL MintPy Mean LOS Velocity over Lucknow's general area (ΔT: 5 years)

The built-up mask overlay (Figure 5.5) confirmed the intersection of subsidence zones with urban infrastructure as a substantial portion of the high-subsidence zones intersects with dense built-up areas. The distinct subsidence hotspots across Lucknow's urban agglomeration are indicative of significant structural stress or hydrogeological imbalances.

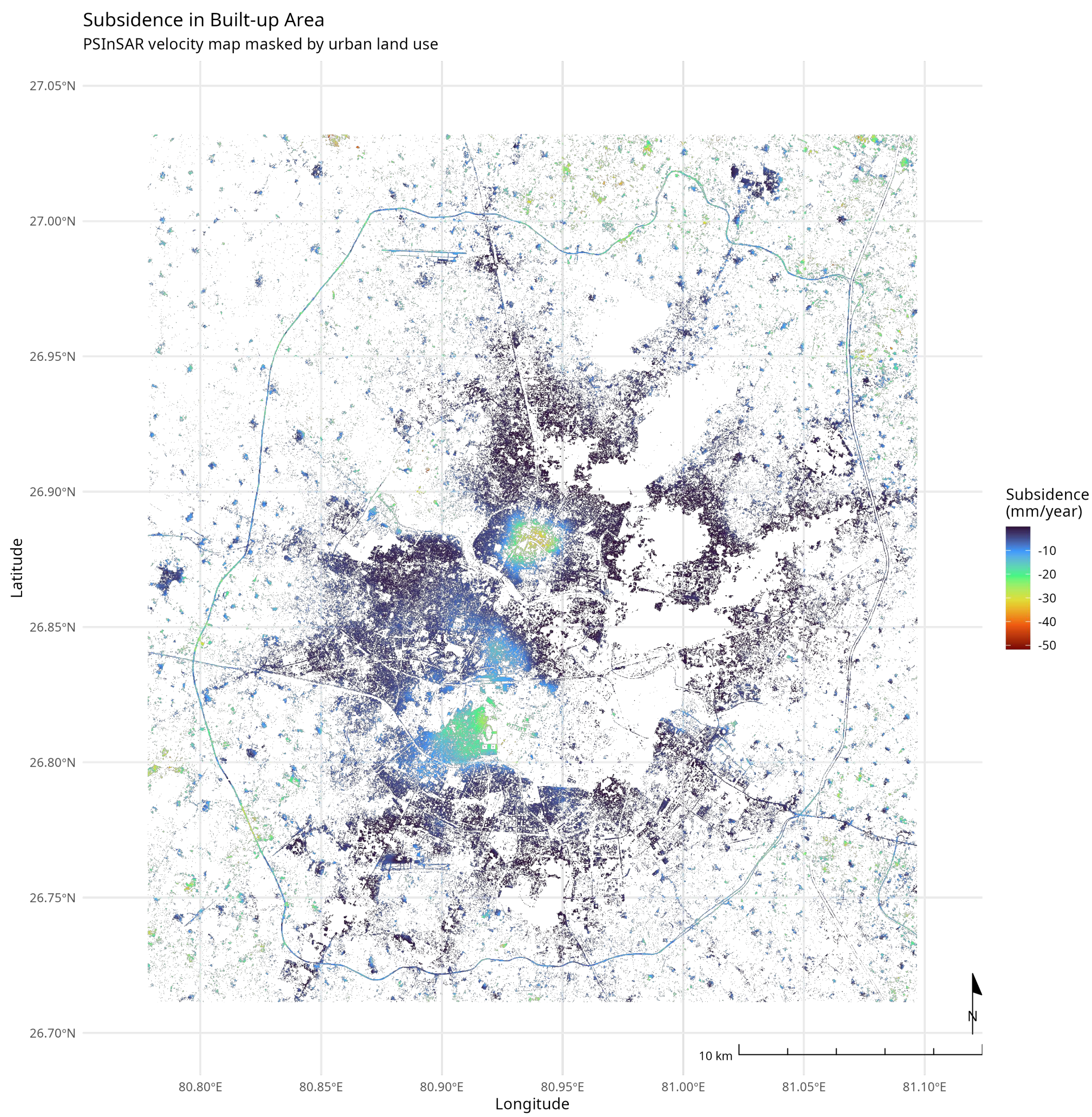


Figure 5.5: OpenSARlab MintPy derived mean LOS Velocity (mm/year) with atmospheric correction over over Lucknow city's built up area observed during 2019/01/04 to 2024/12/27

The land use and land cover (LULC) classification map, shown in Figure 5.6, was generated using Sentinel-2 imagery and a supervised Random Forest classifier.

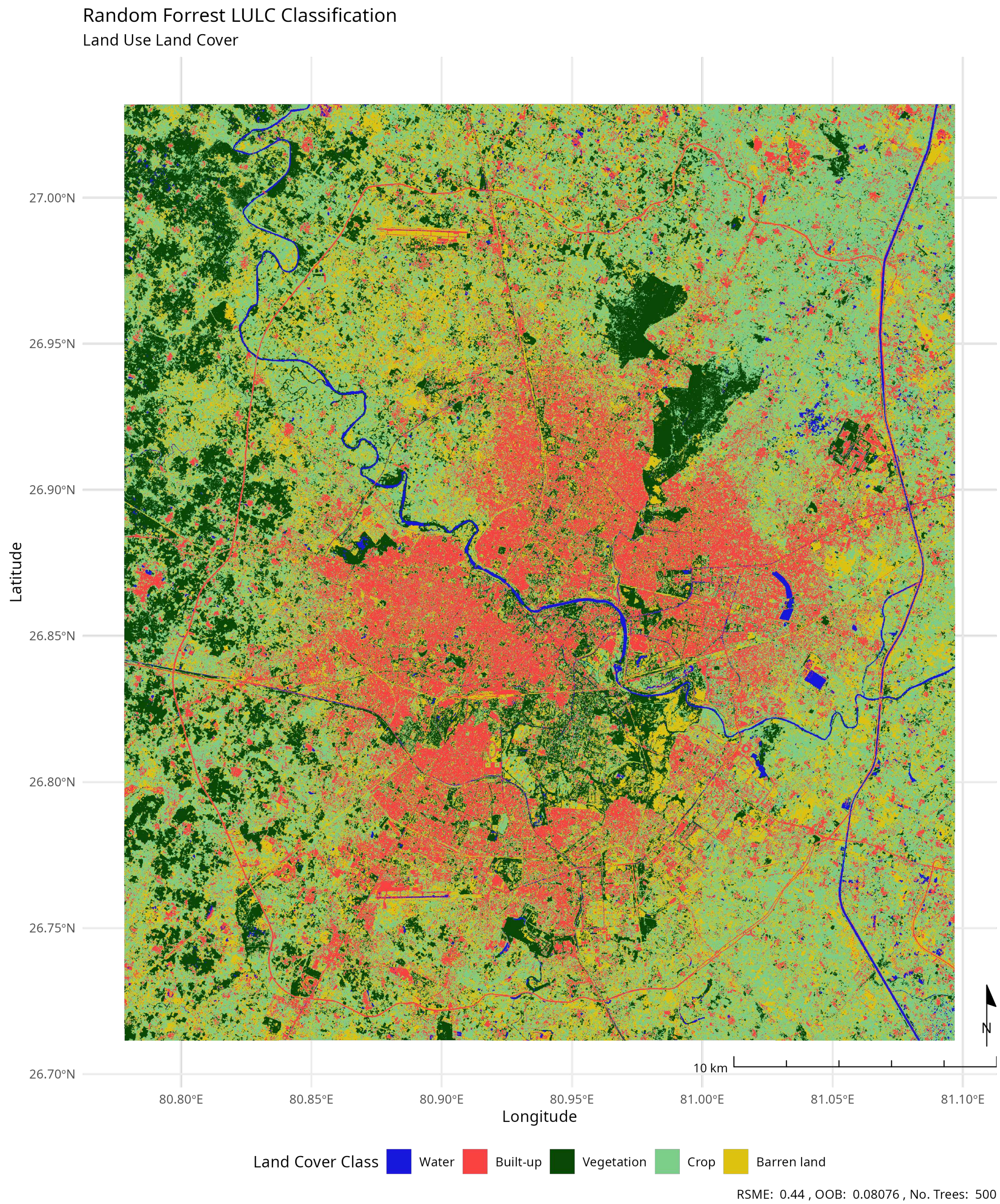


Figure 5.6: Land Use Land Cover (LULC) for the year 2025

The LULC map delineates built-up areas and other categories. This map was used as a critical input for the Land Subsidence Susceptibility Model (LSSM), where each land cover class was encoded as a categorical predictor. Built-up areas and road corridors, identified through LULC and infrastructure layers, emerged as significant factors contributing to subsidence risk due to their association with increased groundwater extraction and anthropogenic load.

Further, the time-series displacement pattern of selected persistent scatterers is illustrated in Figure 5.7.

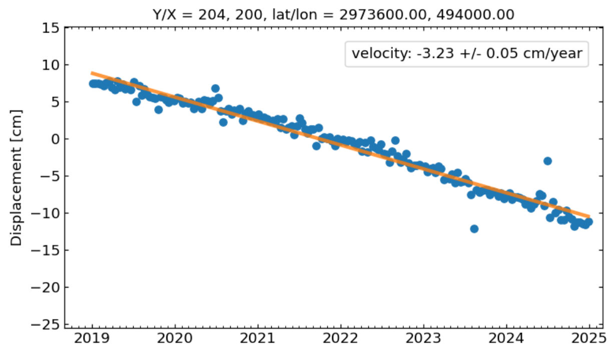


Figure 5.7: Point Displacement Time-series

To ensure the reliability of displacement estimates, atmospheric phase delays caused by stratified tropospheric signals and others were corrected in the MintPy processing workflow (Figure 5.8).

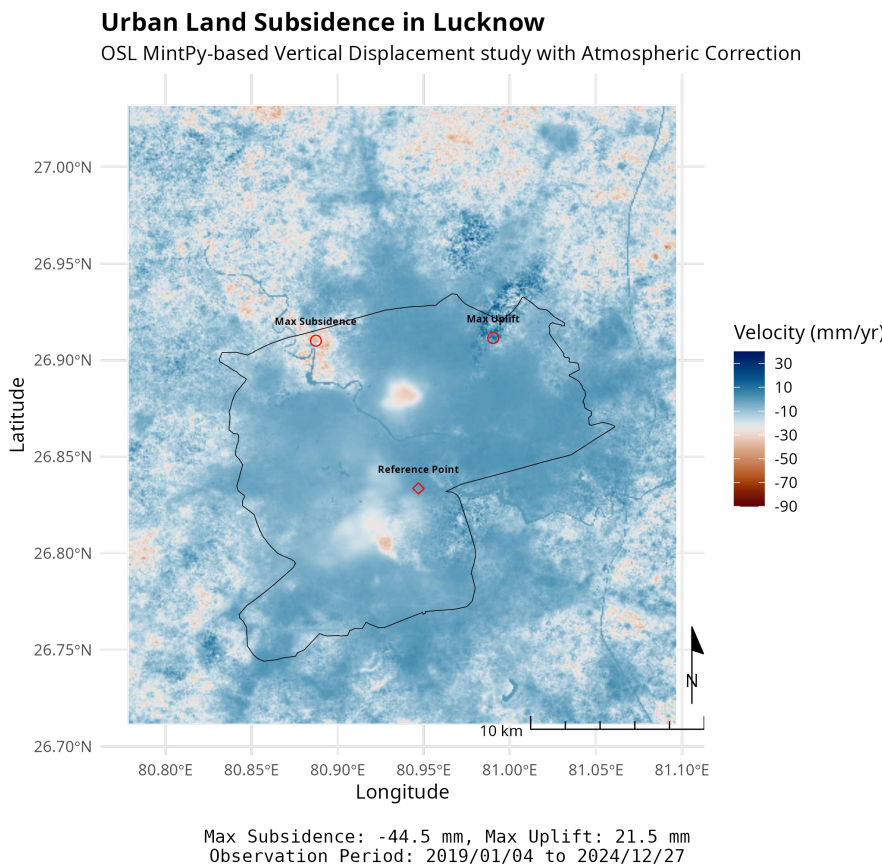


Figure 5.8: Mean LOS Velocity (mm/year) with atmospheric correction over city (ΔT: 5 years)

The difference map (Figure 5.9) highlights zones where uncorrected atmospheric delays significantly influenced the velocity estimates. While not altering the overall spatial trend of subsidence, it reduces local anomalies and enhances phase unwrapping stability.

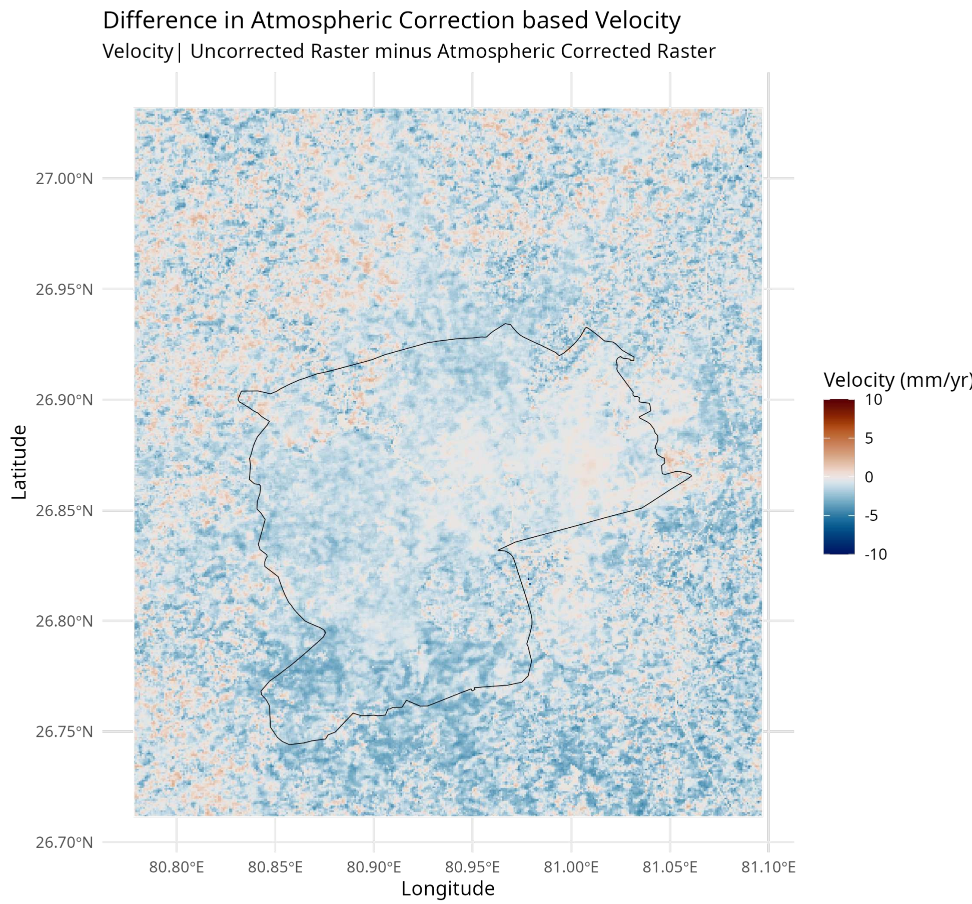


Figure 5.9: Difference in Atmospheric Correction based Velocity

5.3 Validation with SBAS LiCSBAS Outputs

To verify PSInSAR findings, SBAS processing via LiCSBAS was performed on the same Sentinel-1 frame using LiCSAR-derived interferograms. After GACOS atmospheric corrections and SBAS inversion, the results showed consistent deformation patterns. The filtered velocity map (Figure 5.7) demonstrated a strong spatial correlation with MintPy outputs, especially in the two central hotspots.

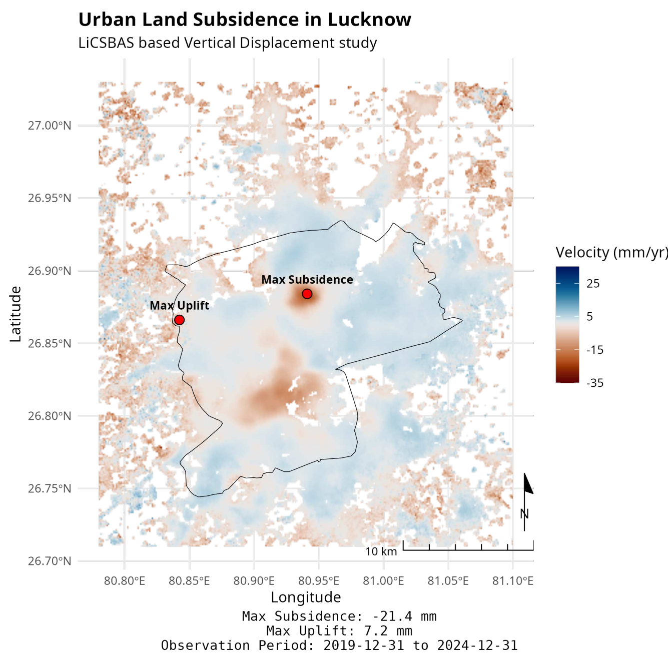


Figure 5.10: LOS Velocity Map from LiCSBAS (ΔT: 5 years)

The linear velocity fit from LiCSBAS indicates an average subsidence rate of around -17.2 mm/year with filtered and unfiltered displacement closely aligned, suggesting high temporal coherence and minimal phase decorrelation at the reference point.

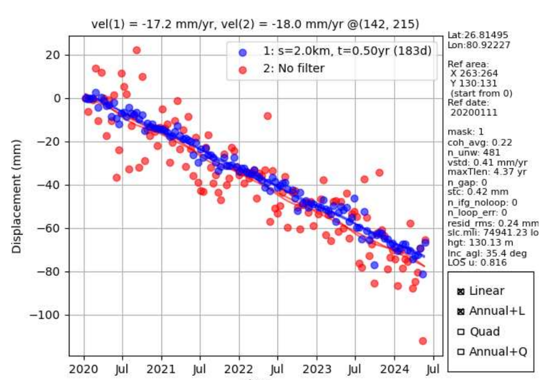


Figure 5.11

Although spatial resolution and pixel density were lower, LiCSBAS confirmed the spatial persistence of subsiding areas. This dual-platform agreement reinforces the reliability of observed trends and affirms the robustness of MintPy's higher-resolution estimates.

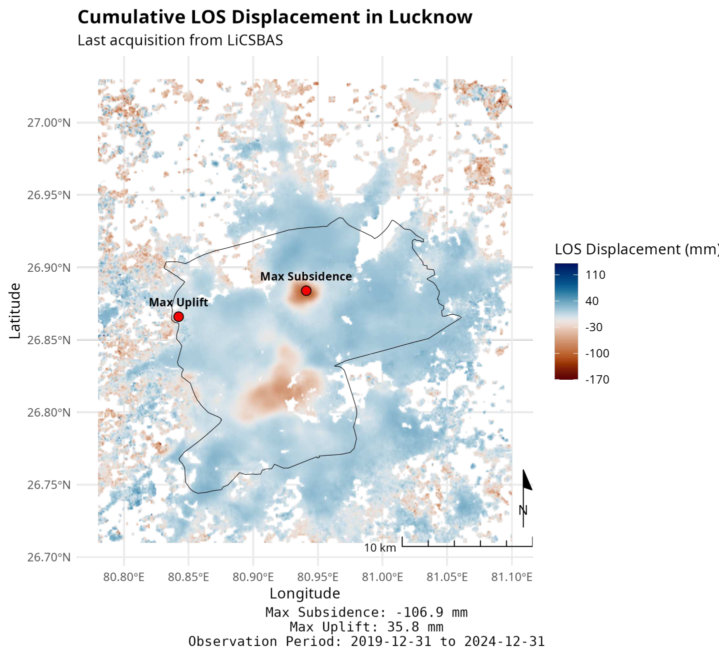


Figure 5.12: LOS Cumulative Displacement Map from LiCSBAS (ΔT: 5 years)

When compared with prior findings (Figure 5.13) based on StaMPS-PSInSAR analysis conducted for the 2014–2020 period, a strong spatial correspondence is observed. Subsidence-prone hotspot detected in this earlier study remain active in the current analysis period (2020–2024).

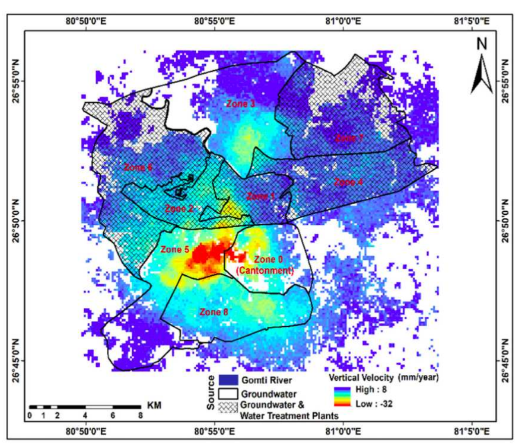


Figure 5.13: Vertical deformation velocity derived using StaMPS based PSInSAR [28]

Atmospheric correction was evaluated within the LiCSBAS workflow using GACOS (Generic Atmospheric Correction Online Service).

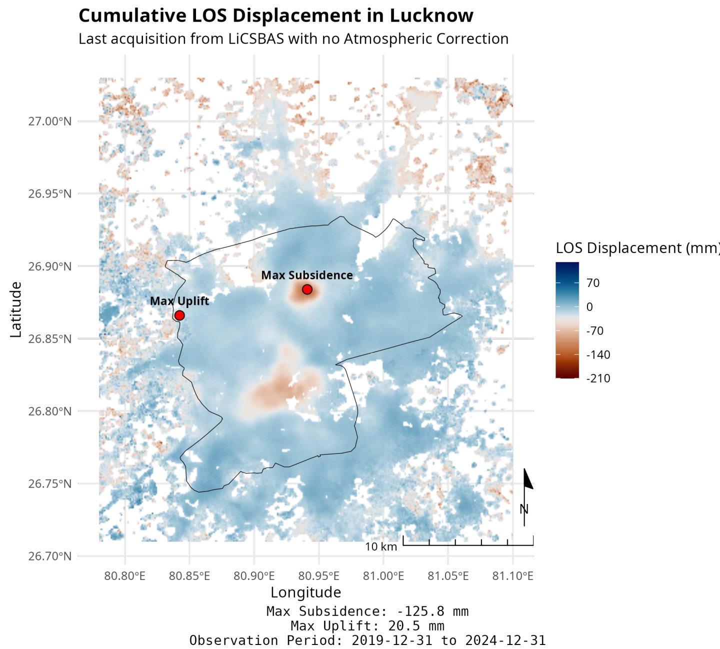


Figure 5.14: LOS Cumulative Displacement Map from LiCSBAS (without GACOS correction) (ΔT: 5 years)

Atmospheric correction helps suppress false signals in the interferogram stack and enhances the interpretability of subsidence trends.

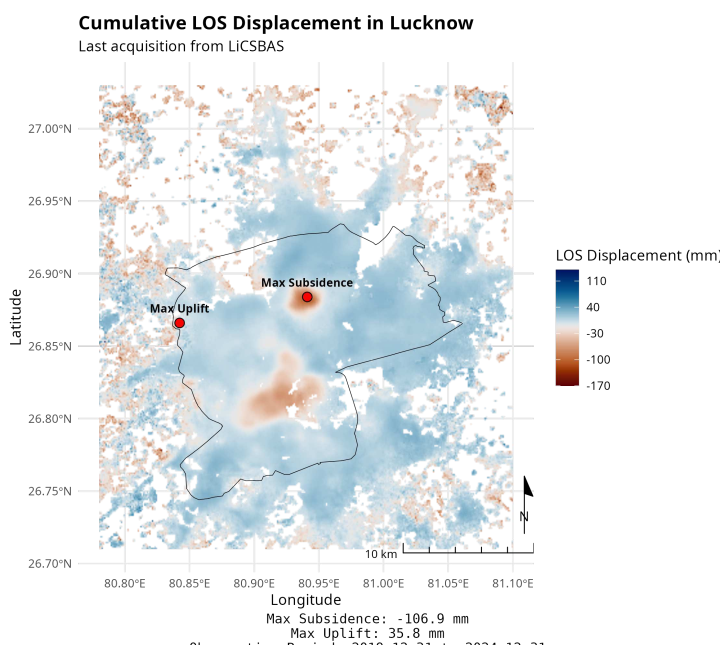


Figure 5.15: LOS Cumulative Displacement Map from LiCSBAS (with GACOS correction) (ΔT: 5 years)

Table 5.1: Key LiCSBAS Parameter Settings for Urban Land Subsidence Mapping

Setting	**Value**	**Typical Range**	**Purpose**
`p01_get_gacos`	y	y/n	Download GACOS atmospheric delay files
`nlook`	1	1–4	Multilook factor to reduce speckle noise
`do03op_GACOS`	y	y/n	Enable GACOS atmospheric correction step
`do04op_mask`	y	y/n	Enable coherence-based masking
`do05op_clip`	y	y/n	Enable geographic clipping of unwrapped phase
`p04_mask_coh_thre`	0.1	0.05–0.3	Coherence threshold for masking unwrapped interferograms
`p12_loop_thre`	1.5	1.0–2.0	Loop closure threshold in radians
`p12_multi_prime`	y	y/n	Use multiple prime images for inversion
`p15_coh_thre`	0.1	0.05–0.3	Threshold to mask low coherence time series pixels
`p15_n_unw_r_thre`	1.0	1.0–1.5	Minimum valid unwrapped ratio per pixel
`p15_stc_thre`	5	2–10	Temporal coherence threshold in mm
`p15_n_ifg_noloop_thre`	30	10–50	Maximum interferograms without loop closure
`p15_n_loop_err_thre`	20	5–50	Maximum loop error threshold
`p15_resid_rms_thre`	5	2–10	Maximum RMS residual threshold in mm
`p16_filtwidth_yr`	0.5	0.3–1.5	Time smoothing window in years
`p16_deg_deramp`	—	1, 2, bl	Polynomial degree of spatial de-ramping
`p16_hgt_linear`	n	y/n	Enable height correlation removal

To ensure accurate and reliable mapping of urban land subsidence, key parameter settings were carefully configured within the LiCSBAS processing workflow (Table 5.1). These settings span critical steps such as multilooking, coherence masking, atmospheric correction using GACOS, loop closure thresholding, and temporal filtering. For instance, a multilook factor of 1 was selected to retain high spatial resolution, while coherence thresholds were applied at multiple stages to suppress noise and preserve reliable interferometric observations.

The adoption of a loop closure threshold of 1.5 radians and multi-prime inversion ensured robustness against phase inconsistencies across the interferogram network. Additionally, temporal filtering and de-ramping options were configured to enhance the interpretability of displacement trends. Collectively, these calibrated settings provided a stable foundation for generating displacement time series and velocity products over urban areas affected by subsidence.

5.4 Key Differences Between LiCSBAS and MintPy Outputs

1. **Interferogram Network Construction LiCSBAS** employs a Small Baseline Subset (SBAS) strategy that selects a clean subset of interferograms and eliminates those failing loop-closure checks.

   MintPy, in contrast, retains a larger set of interferograms, including some with initial inconsistencies. These are corrected using weighted least squares estimation.

   Implication: MintPy may capture more subtle deformation signals, while LiCSBAS prioritizes stability and noise reduction, potentially excluding low-quality but informative interferograms.

2. **Reference Point Selection** Both pipelines use auto-detected reference points; however, they are unlikely to coincide spatially.

   As InSAR measures relative deformation, differing reference points introduce vertical offsets or tilts in the final displacement maps.

   Implication: The absolute deformation values may vary due to reference point discrepancies, even if spatial trends remain similar.

3. **Atmospheric Correction Strategies LiCSBAS** uses GACOS corrections derived from ECMWF data.

   MintPy applies atmospheric phase screen removal using either ERA5 reanalysis data or elevation-based empirical models.

   Implication: The use of different atmospheric models may lead to removal of distinct noise components. In some cases, genuine deformation signals may be dampened or exaggerated depending on the correction approach.

4. **Filtering and Pixel Masking LiCSBAS** applies stricter filtering criteria, masking out pixels with phase unwrapping errors or high noise.

   MintPy retains a larger number of pixels, including those with moderate noise, through phase bridging and temporal correction techniques.

   Implication: MintPy provides broader spatial coverage, whereas LiCSBAS restricts output to high-confidence pixels, resulting in a sparser but potentially more reliable deformation field.

5.5 Land Subsidence Susceptibility Map

The final Land Subsidence Susceptibility Map (LSSM) was generated using an XGBoost classifier trained on PS-InSAR-derived mean LOS velocities and multiple geo-environmental predictors. These included topographic features (slope, elevation, TWI), anthropogenic factors (construction density, road network), and land use categories. The model output which was initially a probability surface was spatially interpolated using Ordinary Kriging to generate a continuous susceptibility surface, which was then classified into five risk zones.

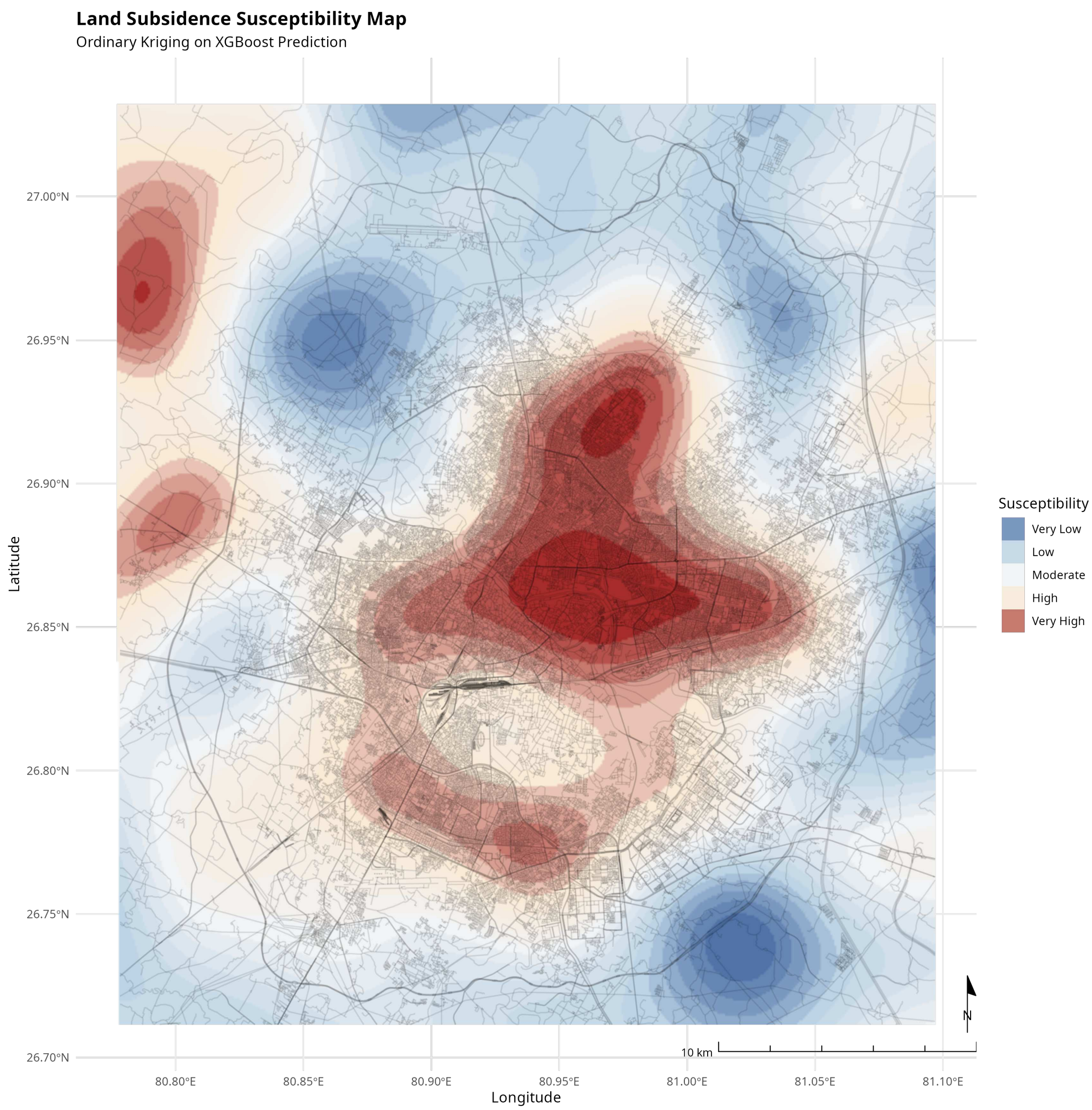


Figure 5.16: Land Subsidence Susceptibility Map (LSSM) — Classified into 5 Risk Zones

High-susceptibility zones strongly overlap with areas experiencing rapid urban expansion and intense groundwater extraction.

5.5.1 Model Evaluation and Feature Importance

The performance of the XGBoost model was evaluated using stratified 10-fold cross-validation. The model achieved a high AUC of 0.91, along with strong F1-score and recall values, indicating robust predictive performance in differentiating subsidence-prone zones.

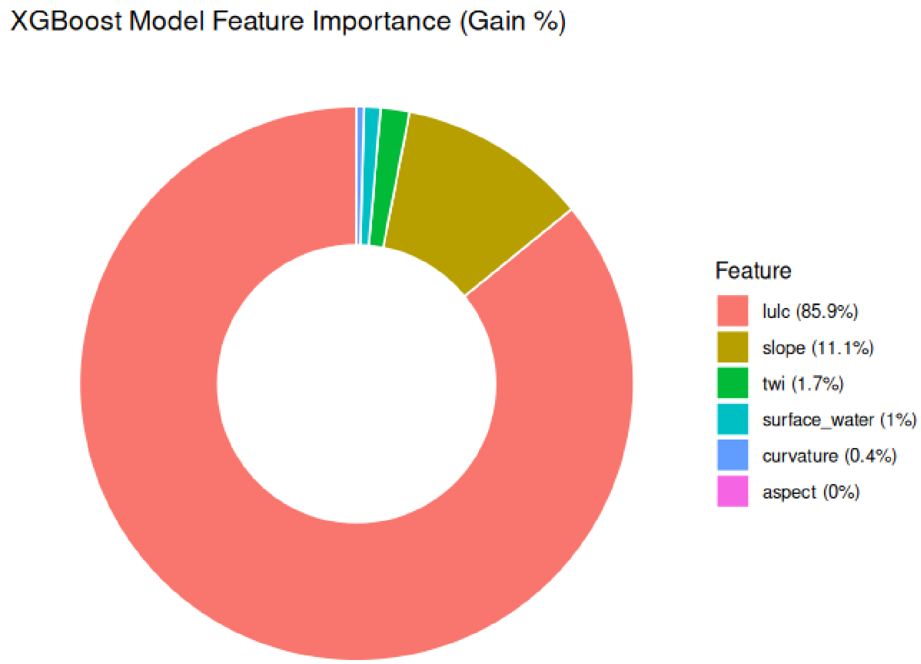


Figure 5.17: Feature Importance Plot from XGBoost Classifier

The most influential features included construction density, slope, road density, and the Topographic Wetness Index (TWI). These results reaffirm the dominance of both anthropogenic pressures and hydrological terrain properties in shaping subsidence vulnerability.

5.6 Model comparison for susceptibility modeling

To ensure robust and generalized subsidence susceptibility modeling, a comparative evaluation of three machine learning classifiers—Random Forest, Support Vector Machine (SVM), and XGBoost—was conducted. Figure 5.18 illustrates the performance of each model across multiple metrics, including Accuracy, AUC, Balanced Accuracy, F1-score, Kappa, Sensitivity, and Specificity.

Among the models, XGBoost achieved the highest scores in most categories, with an AUC of 0.70, Accuracy of 0.69, and F1-score of 0.66. This consistent outperformance established XGBoost as the most suitable model for generating spatial predictions of land subsidence susceptibility.

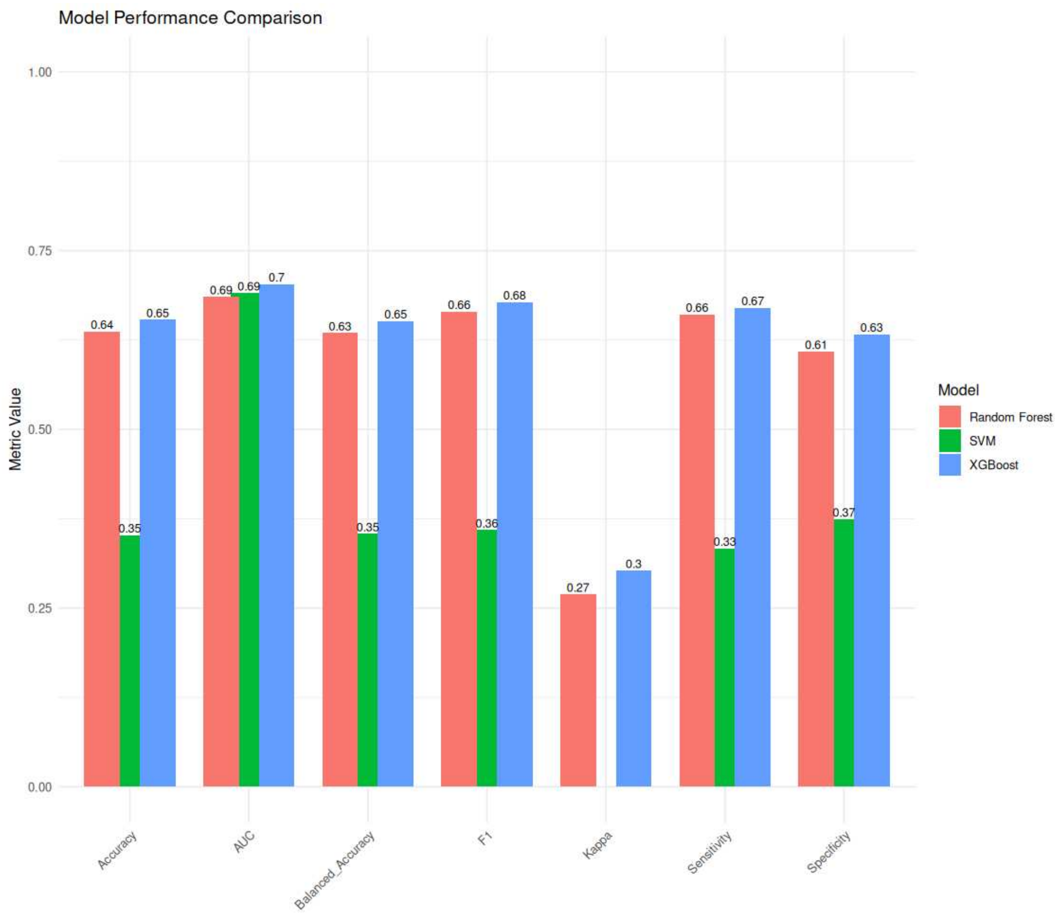


Figure 5.18: Model Performance Comparison: Random Forest, SVM, and XGBoost

5.7 Model Performance

The XGBoost model demonstrated moderate predictive capability given the limited feature set. With an AUC of 0.705, and balanced accuracy of 0.654, the model showed reasonable discrimination between susceptible and non-susceptible zones. Sensitivity (0.673) and overall accuracy (65.6%) further support this performance, while the Kappa statistic (0.308) indicates moderate agreement beyond chance.

Multicollinearity diagnostics (VIF < 2, TOL > 0.5) confirmed that predictors were independent and non-redundant. Incorporating additional datasets—such as soil profile, groundwater levels, or subsurface geology—could improve model accuracy and potentially justify the use of more complex modeling approaches.

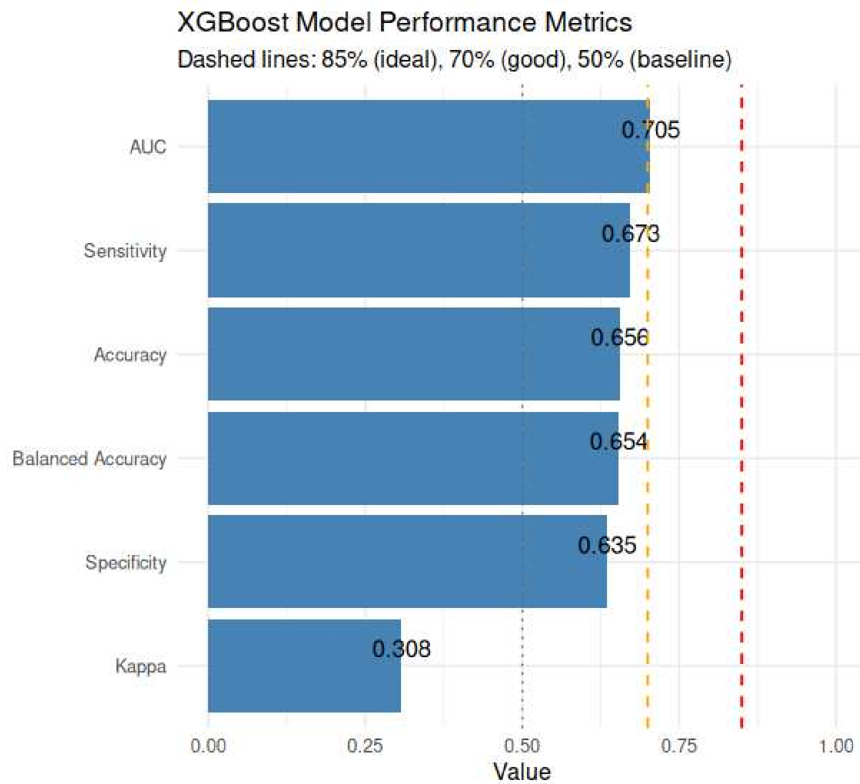
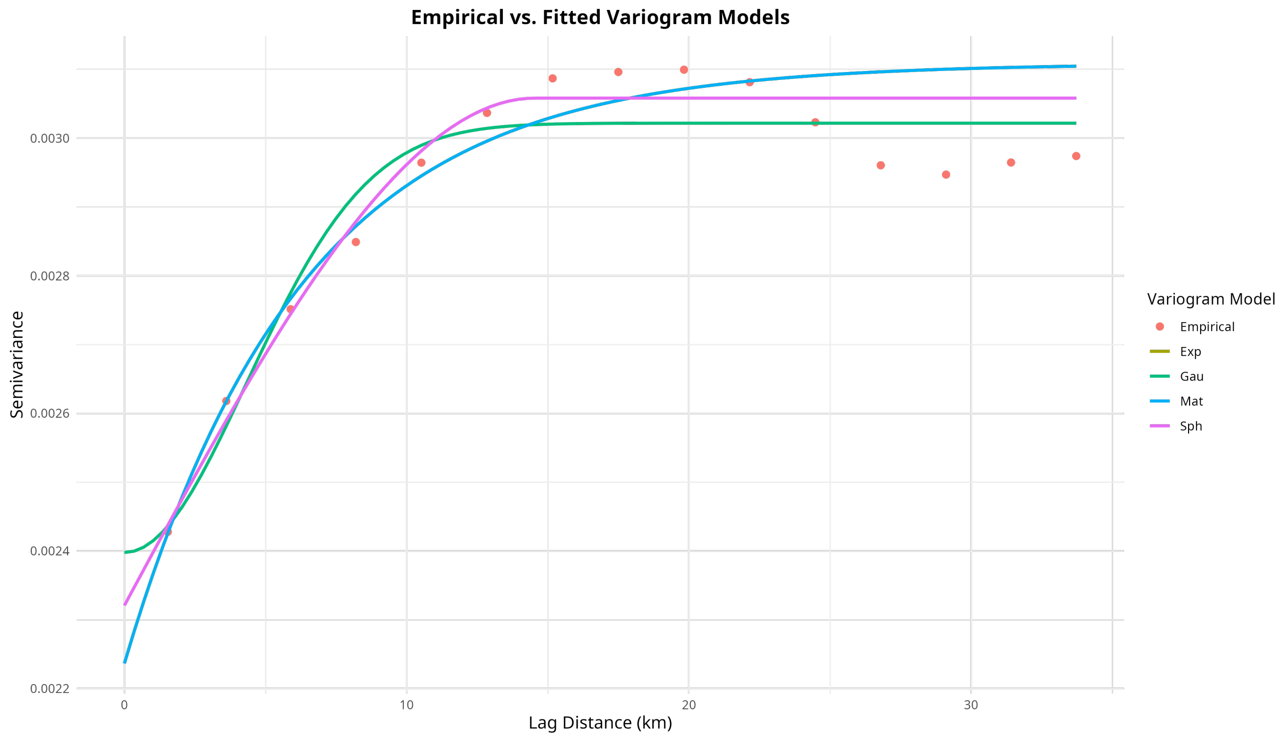


Figure 5.19: XGBoost model performance metrics. AUC, Sensitivity, and Balanced Accuracy exceed the baseline threshold (50%) and approach the good threshold (70%).

Following model selection, XGBoost was used to predict subsidence susceptibility across the entire study region. Figure 5.20 presents the final classified map, highlighting high risk zones predominantly around expanding urban fringes and known groundwater-stressed corridors. These spatial predictions align well with previously observed subsidence zones, reinforcing the validity and spatial consistency of the model's output.

Figure 5.20: XGBoost Model Predictions: Spatial Distribution of Subsidence Susceptibility

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Sph  | Nugget =  0.00232 | Sill =  0.00306 | Range =    14.5353 km | Min Lag =    1.54 km | Max Lag =   33.72 km
Exp  | Nugget =  0.00224 | Sill =  0.00311 | Range =     6.2849 km | Min Lag =    1.54 km | Max Lag =   33.72 km
Gau  | Nugget =  0.00240 | Sill =  0.00302 | Range =     6.1094 km | Min Lag =    1.54 km | Max Lag =   33.72 km
Mat  | Nugget =  0.00224 | Sill =  0.00311 | Range =     6.2849 km | Min Lag =    1.54 km | Max Lag =   33.72 km
```

Figure 5.21: Variogram Model Selection for Ordinary Kriging

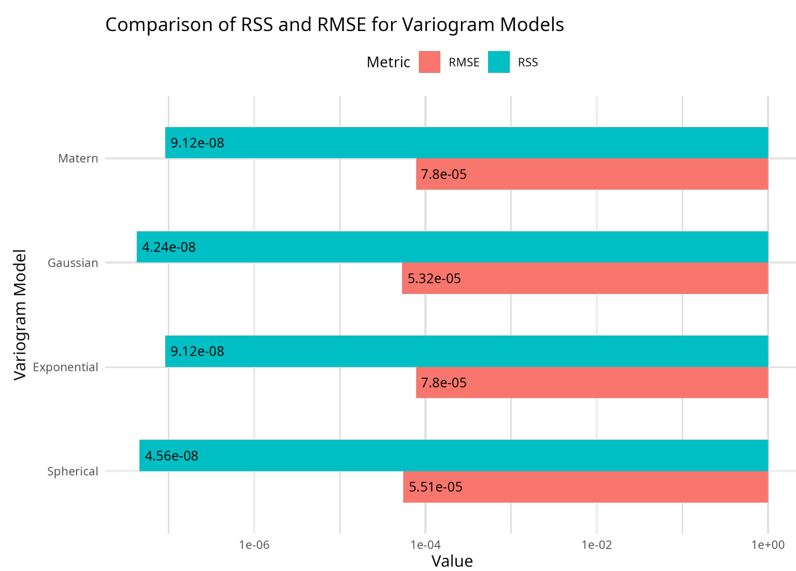

Figure 5.22: RMSE and Residual Sum of Squares (RSS) for Variogram Model Fitting

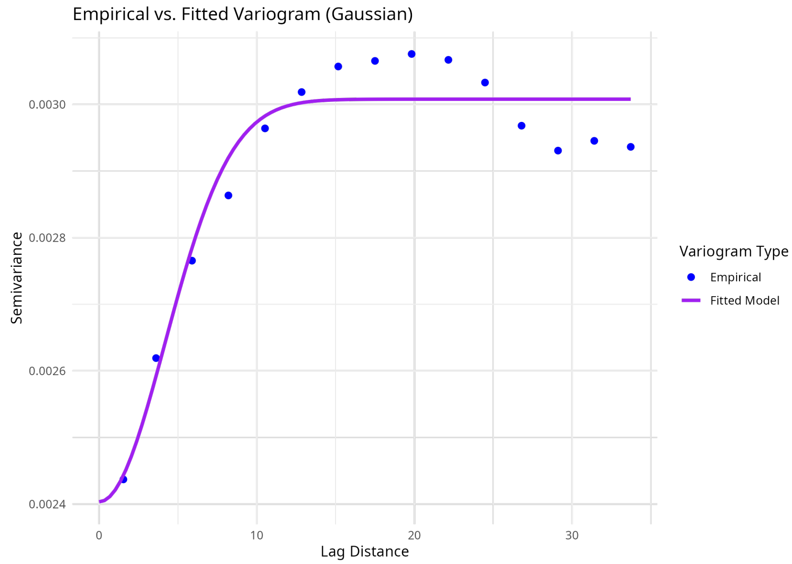

Figure 5.23: Gaussian Variogram Model Fitting

Among the tested variogram models, both Gaussian and Spherical demonstrated close performance. The Gaussian model achieved the lowest RMSE (0.0000000424) and RSS (0.0000532), marginally outperforming the Spherical model, which recorded an RMSE of 0.0000000456 and RSS of 0.0000551. Although the differences were minor, the Gaussian variogram was selected for the final kriging interpolation due to its slightly superior error metrics and its theoretical suitability for modeling smooth, continuous deformation patterns typical of land subsidence phenomena.

5.8 Urban Land Subsidence mapping for selected cities

5.8.1 Delhi

Delhi is a rapidly expanding and has a well documented history of groundwater extraction and infrastructural load. The velocity maps reveal clear patterns of ground deformation. Subsidence hotspot are notably visible in the southern and southwestern areas. Peak deformation rates in localized pockets exceed -45 mm/year.

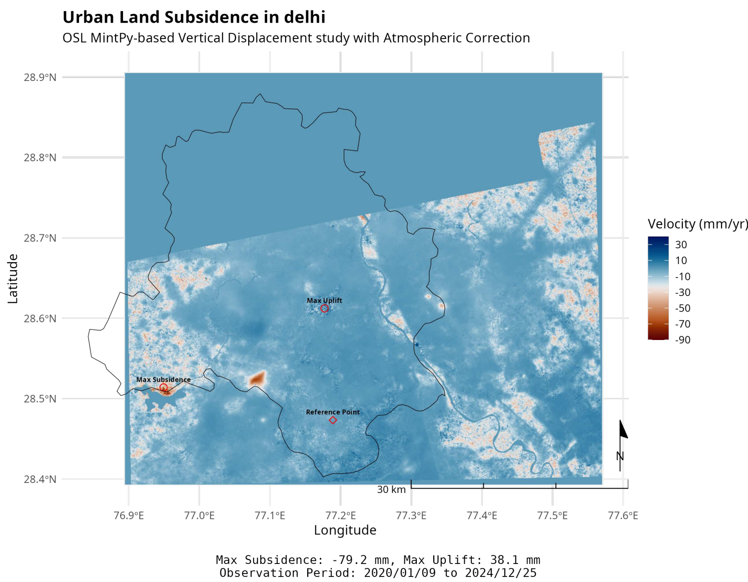

Figure 5.24: OSL MintPy Output for Delhi: LOS Velocity Map (ΔT: 5 years)

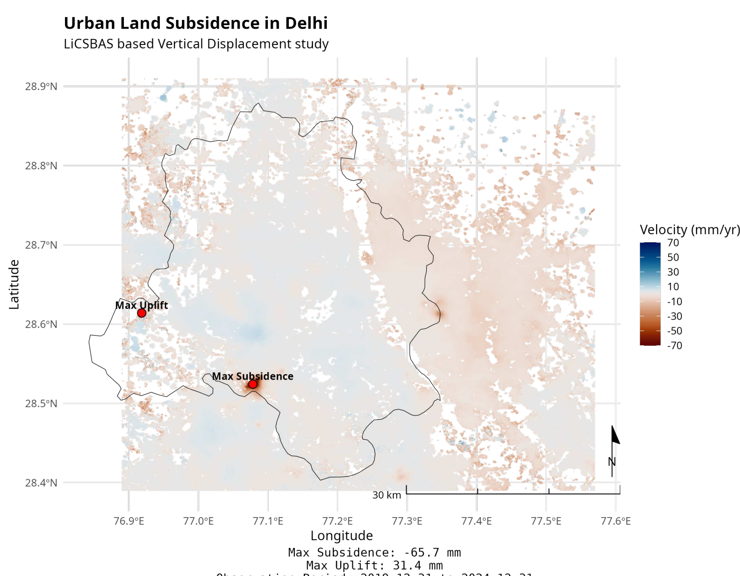

Figure 5.25: LiCSBAS Output for Delhi: LOS Velocity Map (ΔT: 5 years)

5.8.2 Chandigarh

Chandigarh, a planned city, is undergoing localized land deformation not as extensive as in Delhi or Lucknow. The southwestern part of the city shows signs of subsidence with peak deformation rates nearing -20 mm/year inside the city and above -45 mm/year in the peri urban areas.

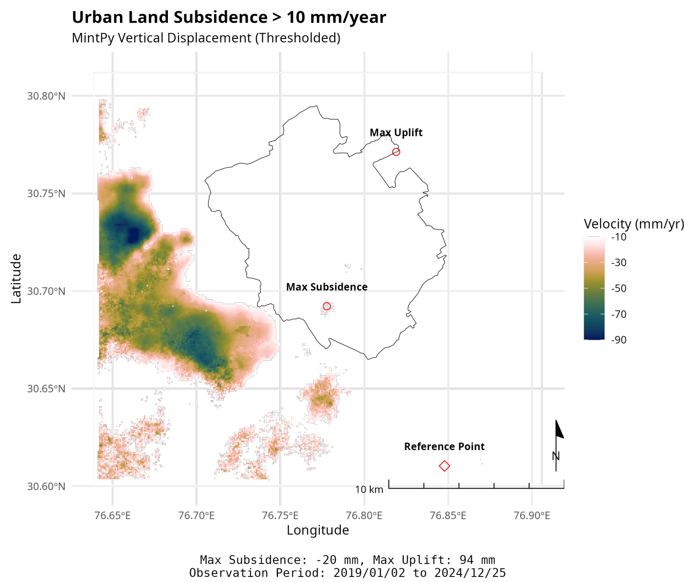

Figure 5.26: OSL MintPy Output: LOS Velocity Map with Thresholding (ΔT: 2.66 years)

SBAS velocity output from LiCSBAS, while the spatial coverage is slightly coarser, shows the overall deformation trend as consistent with the MintPy output.

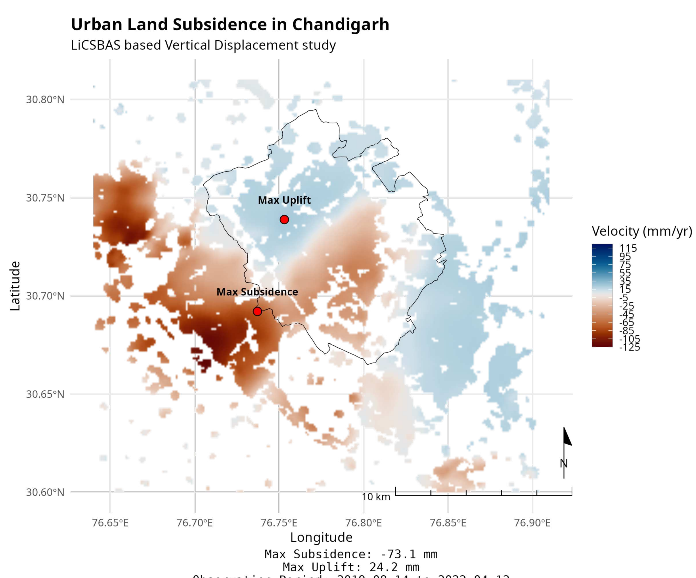

Figure 5.27: LiCSBAS Output: LOS Velocity Map (ΔT: 2.66 years)

5.8.3 Mumbai

Mumbai, a densely populated coastal megacity, is characterized by complex subsurface conditions due to land reclamation, coastal geomorphology, and intense infrastructure loading. The PS-InSAR results derived from MintPy (Figure 5.28) reveal localized subsidence particularly along reclaimed zones and peri-urban edges. Within the city core, deformation rates reach up to -20 mm/year, while certain peripheral areas show subsidence exceeding -45 mm/year. Additionally, minor uplift zones are detected.
Unlike other cities in this study, LiCSBAS did not provide usable SBAS output for Mumbai due to limited spatial coverage and data inconsistency across the selected Sentinel 1 frames. As such, only MintPy outputs were used to analyze deformation dynamics in the Mumbai region.

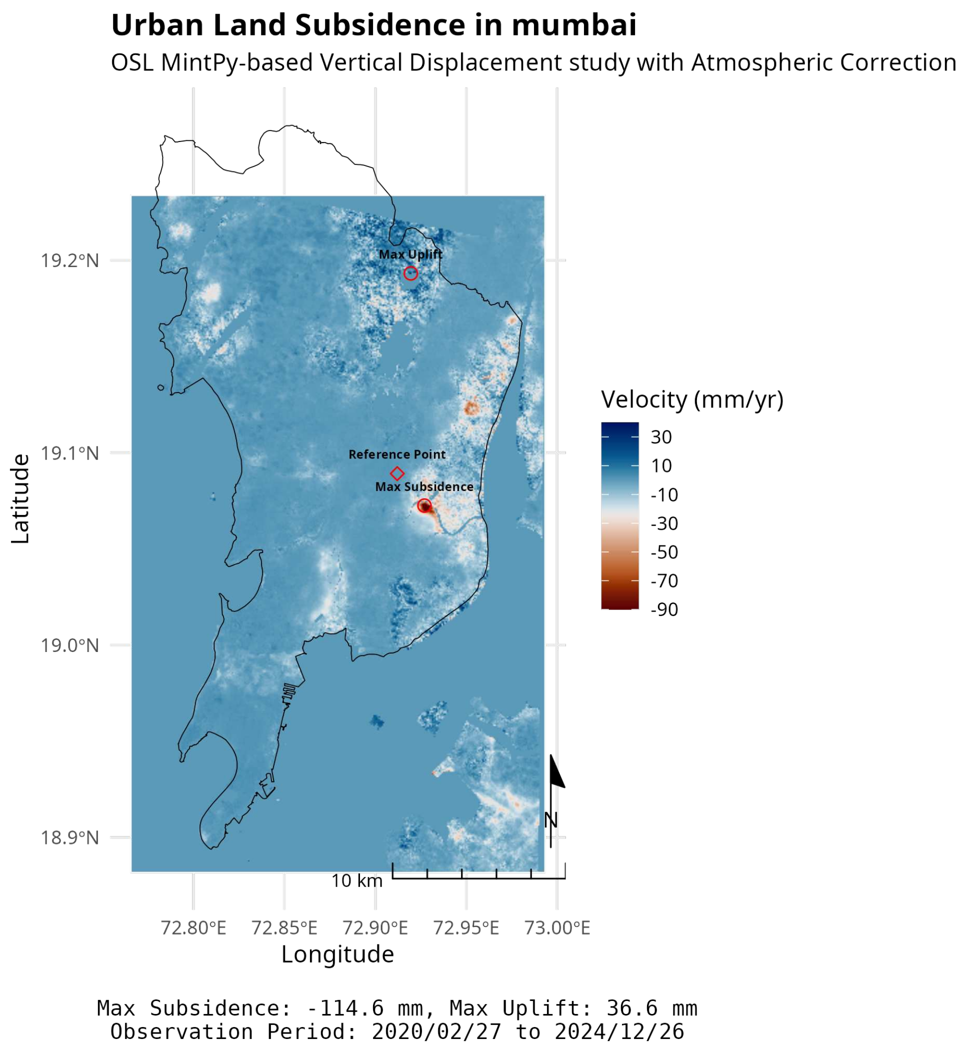

Figure 5.28: MintPy Output for Mumbai: LOS Velocity Map (ΔT: 4.83 years)

5.8.4 Kolkata

Kolkata, a major urban centre in eastern India, faces increasing geotechnical vulnerability due to rapid urbanization, groundwater extraction, and floodplain development. The PSInSAR analysis conducted using MintPy (Figure 5.29) reveals zones of both subsidence and uplift. Notable subsidence is observed in the eastern and southeastern peripheries, with peak deformation rates reaching beyond -90 mm/year. The city core remains relatively stable, while minor uplift trends appear in localized pockets. The reference point was chosen from a stable zone in the northern sector to ensure consistent velocity baseline correction. The subsidence pattern resembles the previous study of the city using InSAR with GPS frameworks to analyse deformation trends during 2007–2011 and 2017–2021 [29].

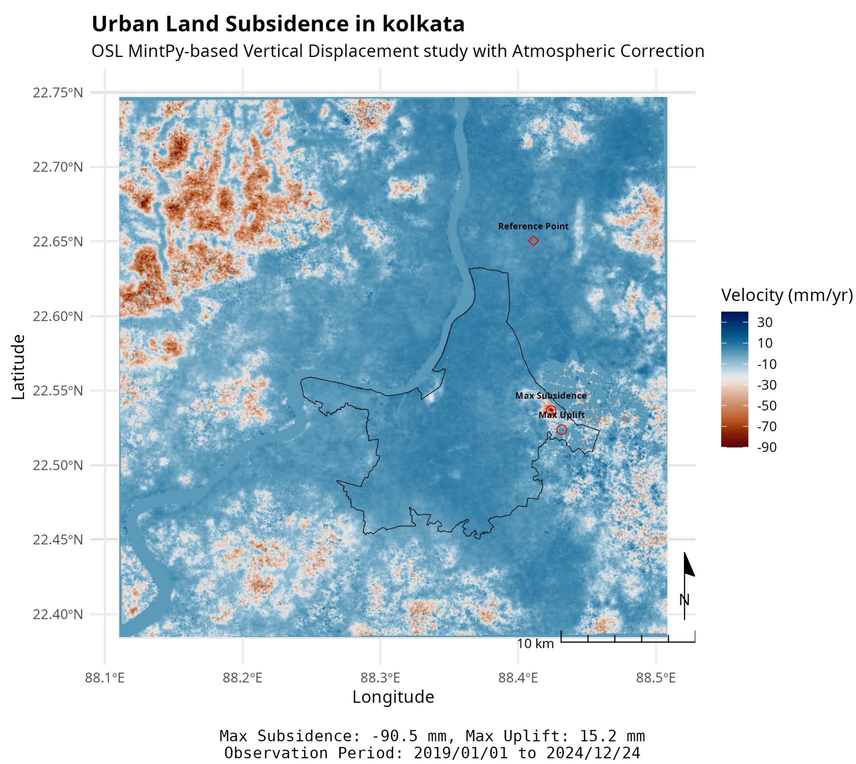

Figure 5.29: MintPy Output for Kolkata: LOS Velocity Map (ΔT: 5.98 years)

Chapter 6

Conclusions

6.1 Key Findings and Methodological Achievements

This study successfully achieved its primary objective: to assess and map land subsidence and susceptibility in an urban context using advanced geospatial and remote sensing techniques. By leveraging time-series analysis of Sentinel-1 SAR data through Persistent Scatterer InSAR (PSInSAR), the research identified and quantified vertical deformation patterns at a fine scale. Using the `MintPy` toolbox, PSInSAR analysis revealed clear subsidence signals in the study area, with certain hotspots exhibiting deformation rates exceeding -25 mm/year, and in extreme cases reaching nearly -87 mm/year. These high-resolution insights offer compelling evidence of ongoing subsurface instability in rapidly urbanizing zones.

To ensure robustness, these results were cross-validated using a second InSAR time-series method; Small Baseline Subset InSAR (SBAS) via the `LiCSBAS` tool. By processing the same Sentinel-1 dataset in parallel and applying GACOS atmospheric corrections, the SBAS outputs confirmed the spatial coherence of deformation patterns found by MintPy. The agreement between both methods—despite using distinct interferogram networks and correction techniques—reinforced the validity of the findings and ruled out methodological artefacts as possible causes of the observed ground motion.

Beyond mapping existing deformation, the study made a contribution through the development of a Land Subsidence Susceptibility Map (LSSM). Here, PSInSAR derived deformation data served as ground truth for training a machine learning model based on Extreme Gradient Boosting (XGBoost). A set of geoenvironmental and anthropogenic predictors were used as inputs. The dataset was carefully prepared, including normalization, synthetic augmentation, and redundancy filtering using Pearson correlation and VIF tests.

The trained XGBoost model demonstrated strong classification performance, achieving an Area Under the Curve (AUC) of approximately 0.705 during cross-validation. This high score indicates the model's ability to differentiate effectively between stable and subsiding areas. Moreover, spatial overlays showed that a large proportion of predicted high-susceptibility zones aligned with observed PSInSAR hotspot, especially those with subsidence rates greater than -10 mm/year. This agreement suggests the model can both replicate known conditions and predict future risk areas with reasonable confidence.

Feature importance analysis further highlighted key subsidence drivers. Construction density emerged as the most influential factor, followed by slope, road density, and the Topographic Wetness Index (TWI). This underscores the prominent role of human-induced stressors, compounded by terrain characteristics, in triggering land subsidence in the study area.

In sum, the study mapped ongoing subsidence using InSAR techniques and developed a data-driven predictive framework to assess susceptibility. This combination of retrospective mapping and forward-looking risk assessment offers a powerful tool for urban hazard management.

6.2 Implications for Urban Infrastructure, Planning, and Policy

The spatial alignment of subsidence hotspot with rapidly urbanizing areas has serious implications for infrastructure resilience and land management. High-susceptibility zones identified in the LSSM correspond to new settlements, industrial zones, and major transportation corridors, areas experiencing high construction loads and groundwater stress. These findings indicate that vital infrastructure in these zones is at increased risk of structural instability.

The LSSM serves as a practical tool for urban planners and policymakers. It enables targeted mitigation strategies, such as foundation reinforcement, land-use zoning, drainage improvements, or groundwater extraction limits. By providing spatially explicit risk information, the study supports more informed urban development and sustainable groundwater governance. Regulatory agencies can use these insights to enforce building codes in subsidence-prone areas and prioritize monitoring where it's most needed.

6.3 Future Work and Recommendations

To build on this framework, future research should integrate physical groundwater and geotechnical models with the current InSAR-ML pipeline. Linking observed deformation with aquifer dynamics would improve causal understanding and enable predictive scenario analysis. Additionally, applying this scalable workflow to other urban areas will help generalize its effectiveness across different geographies.

Further technical enhancements, such as advanced atmospheric correction, GNSS-based validation, or machine learning filters for noise reductio, could improve sensitivity and accuracy. Integrating these developments will enhance the model's performance and ensure that land subsidence monitoring remains a critical component of resilient, data-informed urban planning.

6.4 Final Remarks

This research presents a comprehensive, scalable, and scientifically validated approach for urban land subsidence monitoring and risk mapping. By combining the strengths of PS-InSAR and SBAS time-series analysis with advanced machine learning and feature interpretation, it not only identifies current subsiding areas but also provides a forward-looking framework to anticipate future risk. The study has immediate applications for urban planning, infrastructure design, and groundwater policy, and it lays the groundwork for future interdisciplinary research at the intersection of Earth observation, data science, and urban sustainability.

As cities around the world continue to expand and confront increasing environmental pressures, including land subsidence, methodologies like the one developed here will be essential for building resilience. By aligning scientific innovation with policy relevance, this work contributes meaningfully to the broader goal of sustainable and adaptive urban development.

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